

Install & Configure AWS CLI

[Edition 05]

[Last Update 240111]

Scenario:

Setting Up a Development Environment on AWS

You work for a software development team, and your company has decided to move its development workflow to AWS for better scalability and collaboration. Your team needs to set up a standardized development environment on Windows EC2 instances to ensure consistency across all developers.

Description:

In this AWS lab, we will shift to AWS for improved collaboration. To set up a standardized environment on Windows EC2 instances, install AWS CLI version 2 for seamless command-line interaction. Configuring AWS CLI credentials to streamline version control, ensuring an efficient, consistent development experience on AWS for the team.

For any issues/help contact in Training WhatsApp Group or Support Ticketing System

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1 INTRODUCTION

The AWS Command Line Interface (AWS CLI) is an open-source tool that enables you to interact with AWS services using commands in your command-line shell. With minimal configuration, the AWS CLI enables you to start running commands that implement functionality equivalent to that provided by the browser-based AWS Management Console from the command prompt in your favorite terminal program:

Windows command line – On Windows, run commands at the Windows command prompt.

Remotely – Run commands on Amazon Elastic Compute Cloud (Amazon EC2) instances through a remote terminal program such as PuTTY or SSH, or with AWS Systems Manager.

The AWS CLI is available in two versions and the information in this guide applies to version 2.x unless stated otherwise:

- **Version 2.x** – The current version, generally available release of the AWS CLI that is intended for use in production environments. This version does include some "breaking" changes from version 1 that might require you to change your scripts so that they continue to operate as you expect.
- **Version 1.x** – The previous version of the AWS CLI that is available for backwards compatibility.

Architecture Diagram:



This activity guide cover steps for:

1. Installing AWS CLI On Windows EC2 Instance
2. Configuring AWS CLI
3. Create S3 Bucket using AWS CLI
4. Terminate Resources

2 OVERVIEW & KEY CONCEPTS

In this section, we'll cover key concepts used in this guide, including an overview of what's covered in this guide.

2.1 What is EC2?

Amazon Elastic Compute Cloud (Amazon EC2) delivers scalable computing power on-demand in the Amazon Web Services (AWS) Cloud. Using Amazon EC2 lowers hardware expenses, allowing you to create and deploy apps quickly. Amazon EC2 allows you to deploy as many or as few virtual servers as you need, set security and networking, and manage storage. You can add capacity (scale up) to address compute-intensive operations such as monthly or annual procedures or surges in website traffic. When consumption falls, you can lower capacity.

2.2 AWS CLI

The AWS Command Line Interface (AWS CLI) is a unified tool to manage your AWS services. With just one tool to download and configure, you can control multiple AWS services from the command line and automate them through scripts.

The AWS Command Line Interface (AWS CLI) is a free and open-source tool that allows you to interface with AWS services using command-line shell instructions. The AWS CLI allows you to start executing commands that implement functionality equal to that given by the browser-based AWS Management Console from the command prompt in your terminal program with little configuration:

Linux shells –

To run commands in Linux or macOS, use typical shell programs such as bash, zsh, and tcsh.

Windows command line –

On Windows, use the command prompt or PowerShell to execute commands.

Remotely - Execute commands on Amazon Elastic Compute Cloud (Amazon EC2) instances via a remote terminal program like PuTTY or SSH, or using AWS Systems Manager.

The AWS API and AWS CLI provide access to all IaaS (infrastructure as a service) and AWS administration, management, and access capabilities offered in the AWS Management Console. At launch or within 180 days after introduction, new AWS IaaS products and services give complete AWS Management Console capability through the API and CLI.

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3 DOCUMENTATION LINKS

1. What Is the AWS Command Line Interface?

<https://docs.aws.amazon.com/cli/latest/userguide/cli-chap-welcome.html>

2. How to create AWS Access Keys

<https://aws.amazon.com/premiumsupport/knowledge-center/create-access-key/>

3. Managing Access Keys for IAM Users

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_access-keys.html#Using_CreateAccessKey

4. Creating an IAM User in Your AWS Account

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_users_create.html#id_users_create_console

5. Developing with Amazon Web Services

<https://aws.amazon.com/developers/getting-started/>

4 PRE-REQUISITE

1. An AWS Account (Free or Paid).

2. A Windows EC2 Instance

Note: For this lab we will require to launch a Windows Virtual Machine Using EC2. If you do not know how to create please check the following lab.

Activity Guide (Lab): Create And Connect To Windows EC2 Instance

3. For Mac users we recommend you use Windows Virtual Machine and the steps for it are shared in the above link.

5 INSTALLING AWS CLI

AWS CLI gives you the ability to automate the entire process of controlling and managing AWS services through scripts. These scripts make it easy for users to fully automate cloud infrastructure.

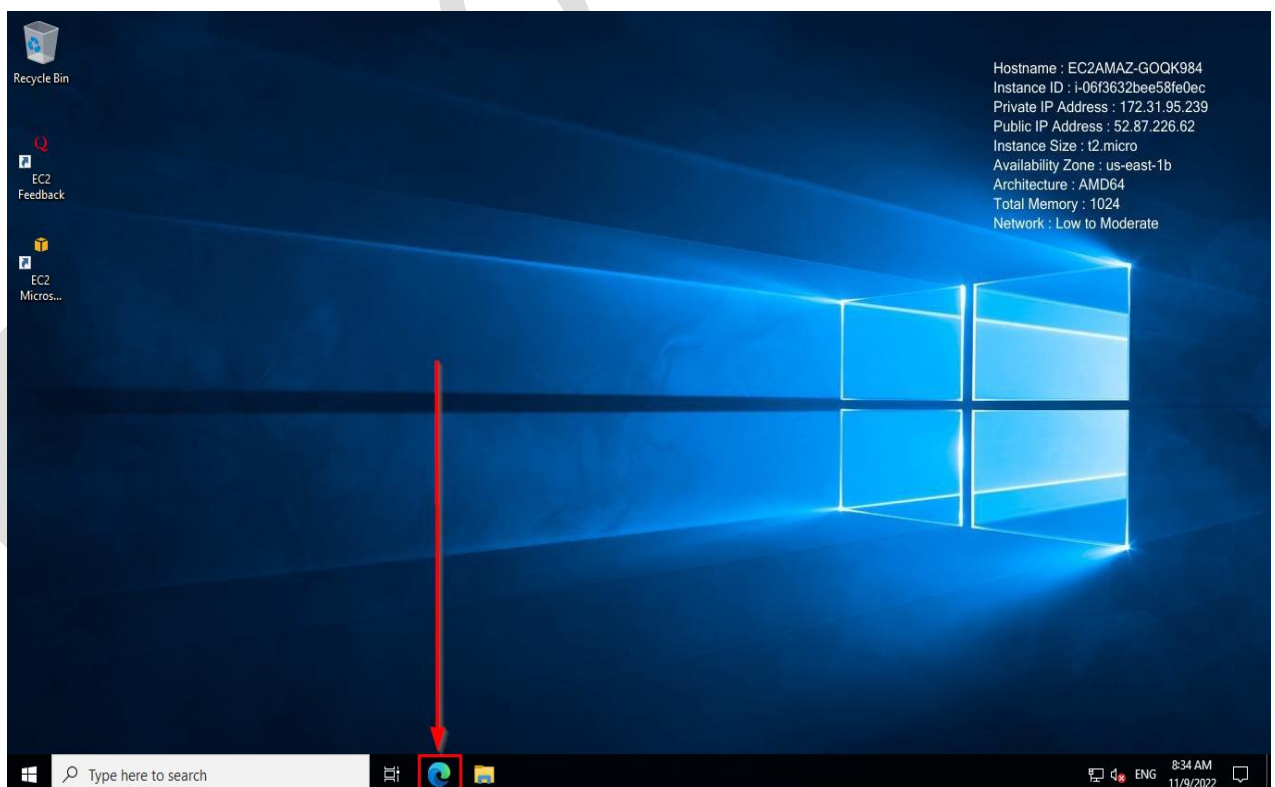
In this section, we will install AWS CLI on Windows EC2 Instance

5.1 On Windows

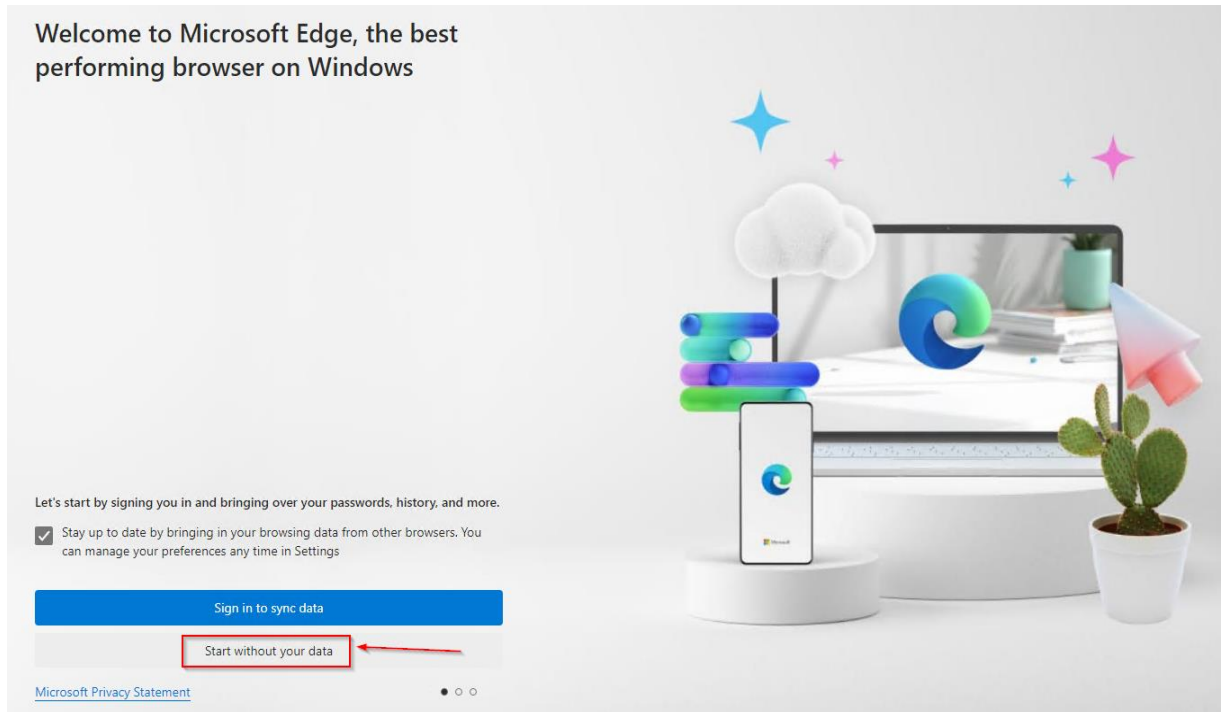
We will be installing AWS CLI version 2.x, the current version intended for use in production environments.

Official Documentation : <https://docs.aws.amazon.com/cli/latest/userguide/install-cliv2-windows.html>

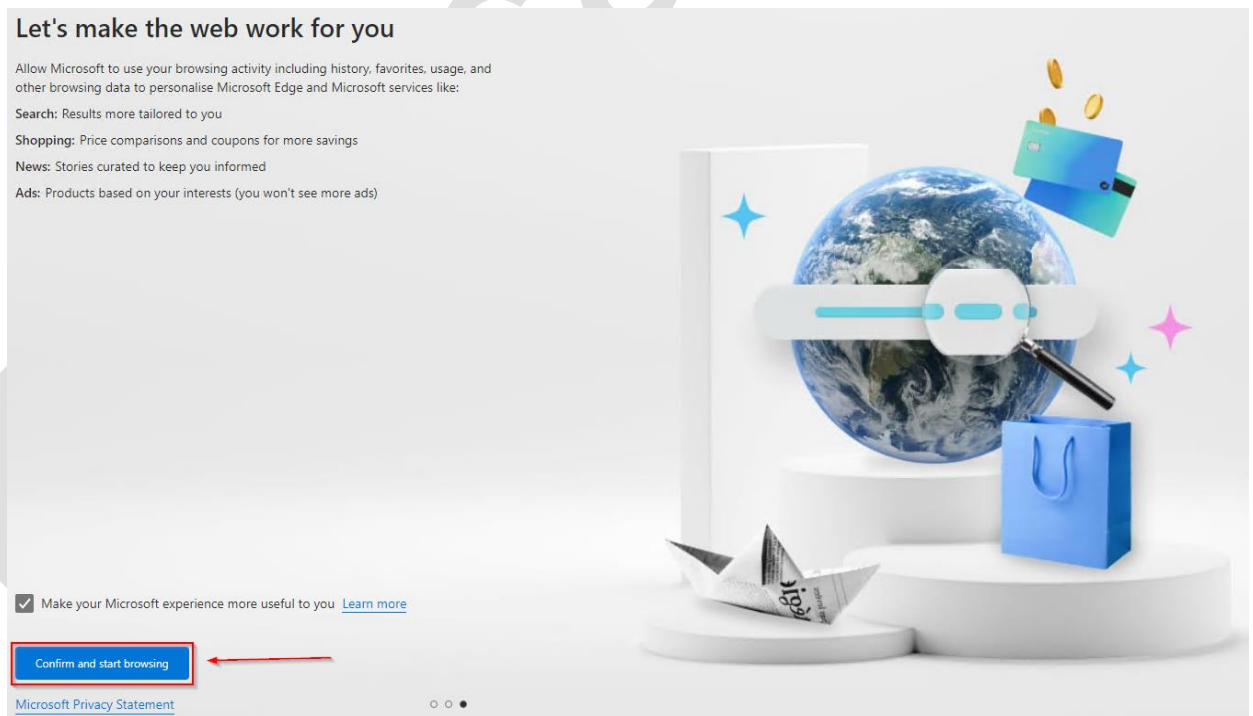
1. Connect to your EC2 Windows instance. If you don't know how to connect EC2 Windows Instance, then check our activity guide on **Activity Guide (Lab): Create And Connect To Windows EC2 Instance**
<https://k21academy.com/topic/activity-guide-creating-a-windows-ec2-instance-2/>
2. Once you have launched the windows EC2 Instance your screen shall look like this then click on **Microsoft Edge** and open it



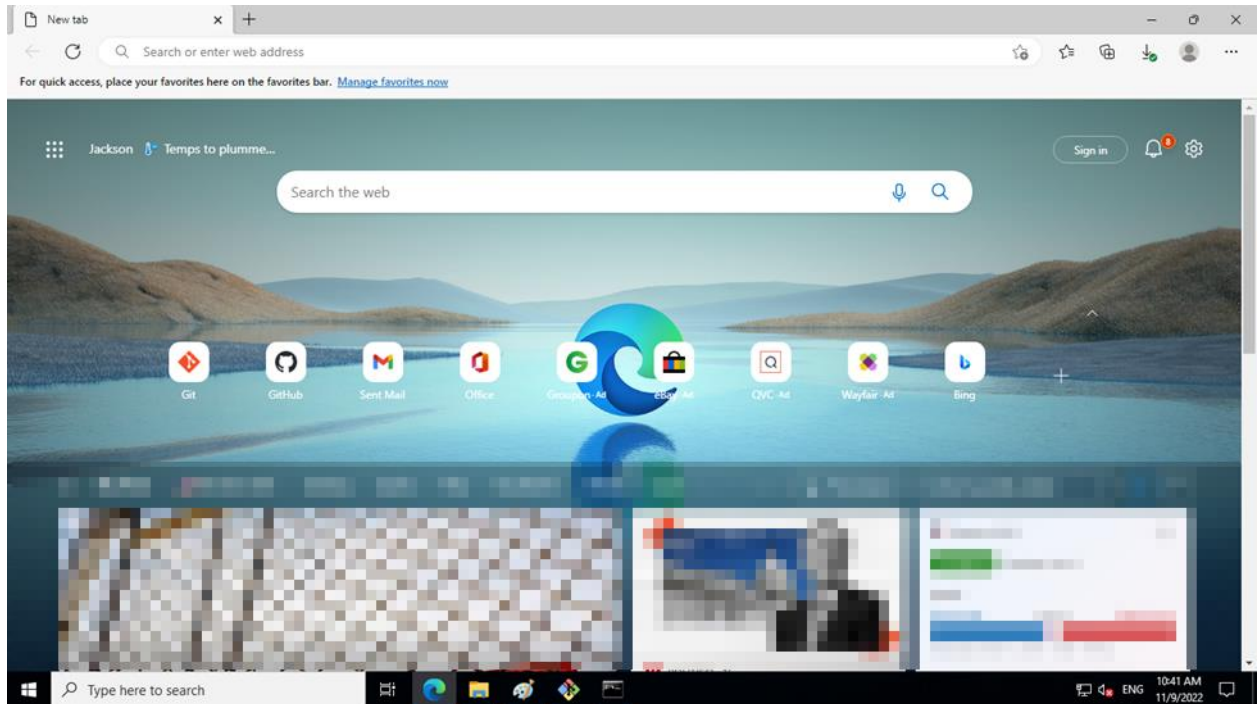
3. Now Click on **Start without your data.**



4. Click on **Confirm and start browsing.**



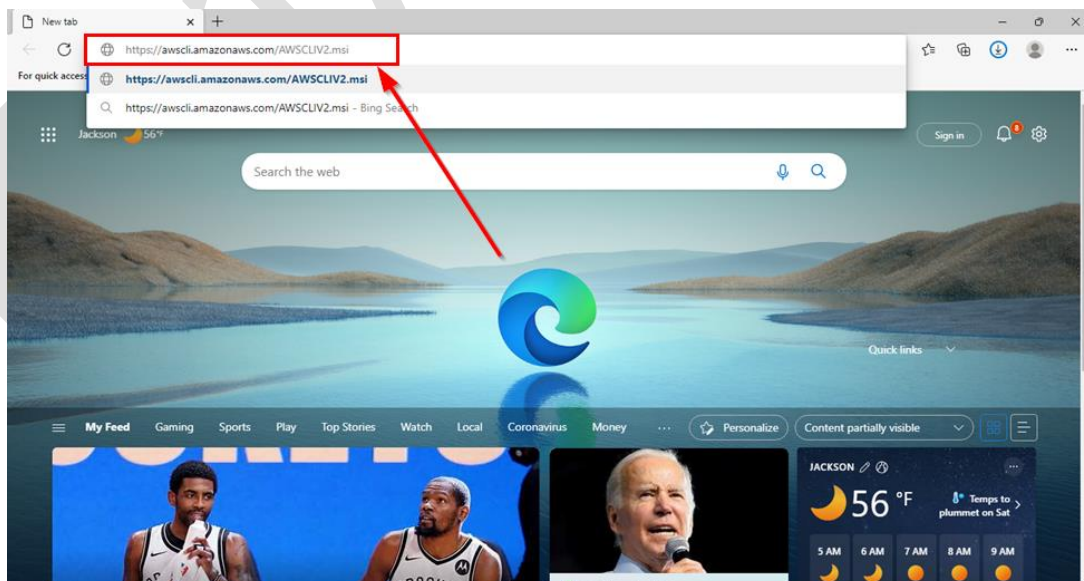
5. It will open a new tab for you.



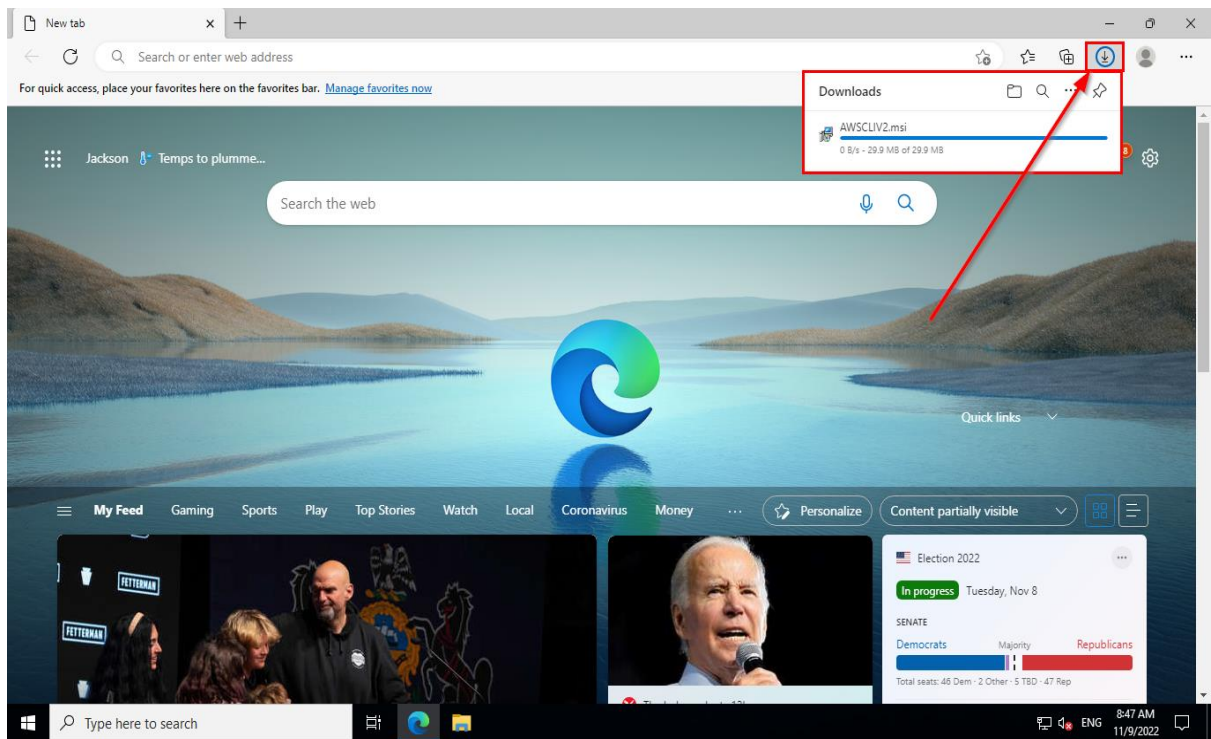
6. Copy the link and hit into your browser inside your Windows Virtual Machine. Download the AWS CLI MSI installer for Windows (64-bit)

<https://awscli.amazonaws.com/AWSCLIV2.msi>

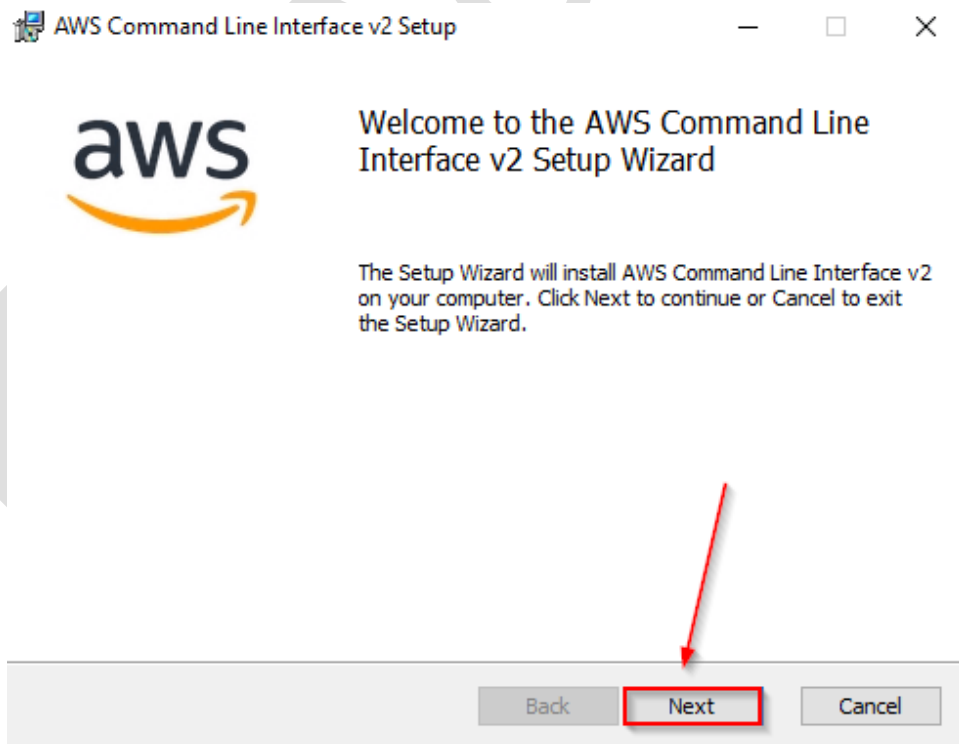
Note: In Future to update your current installation of AWS CLI version 2 on Windows, download a new installer each time you update to overwrite previous versions. AWS CLI is updated regularly, so check the [Releases page on GitHub](#) to see when the latest version was released.



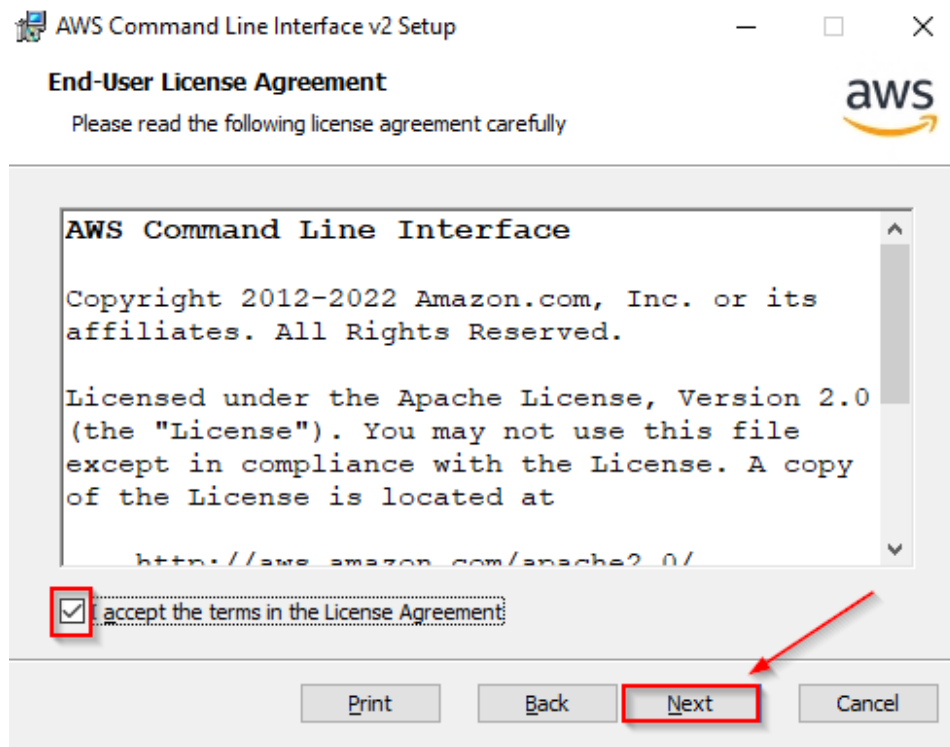
7. When it starts downloading, it appears under the downloads icon. Once the file is downloaded, you can open it directly by clicking on it.,



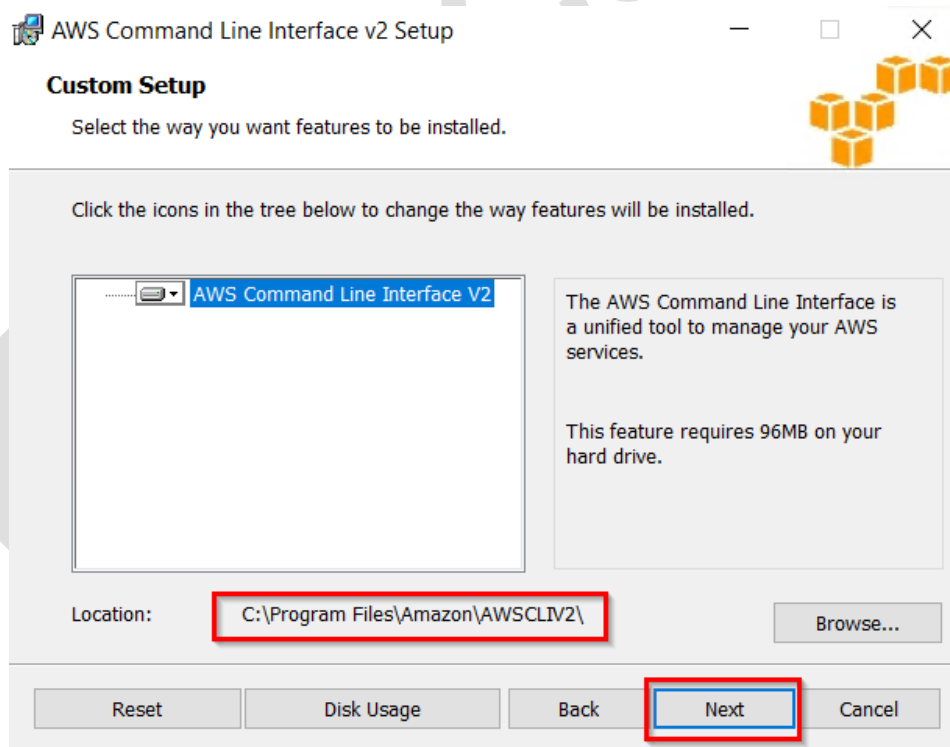
8. Now Click on **Next**.



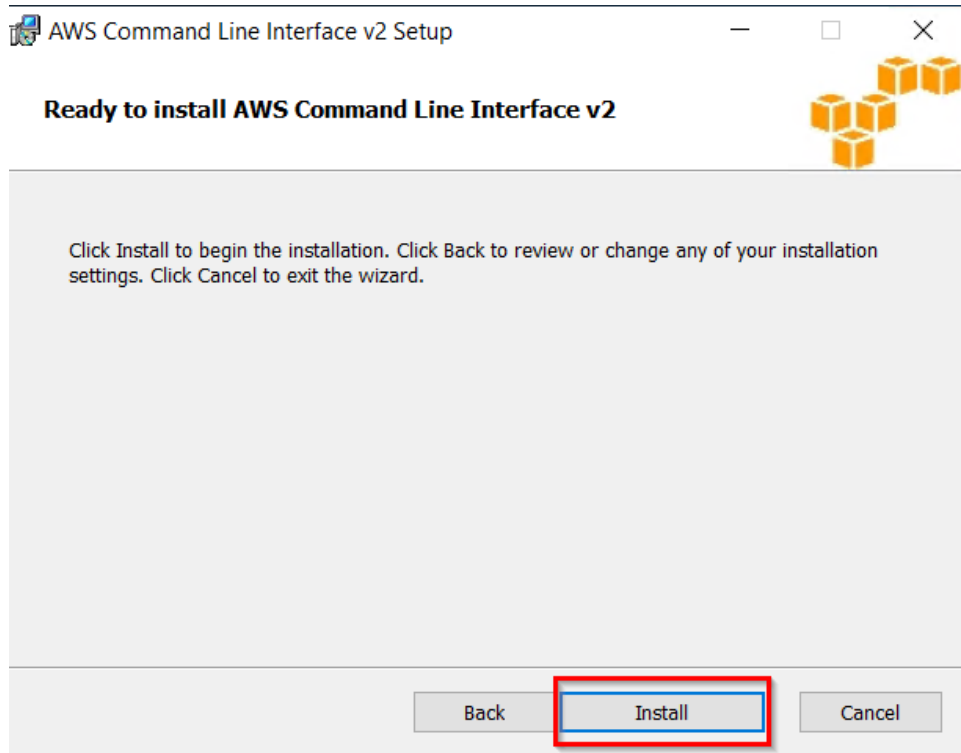
9. Tick the check box of license agreement and click on **Next**.



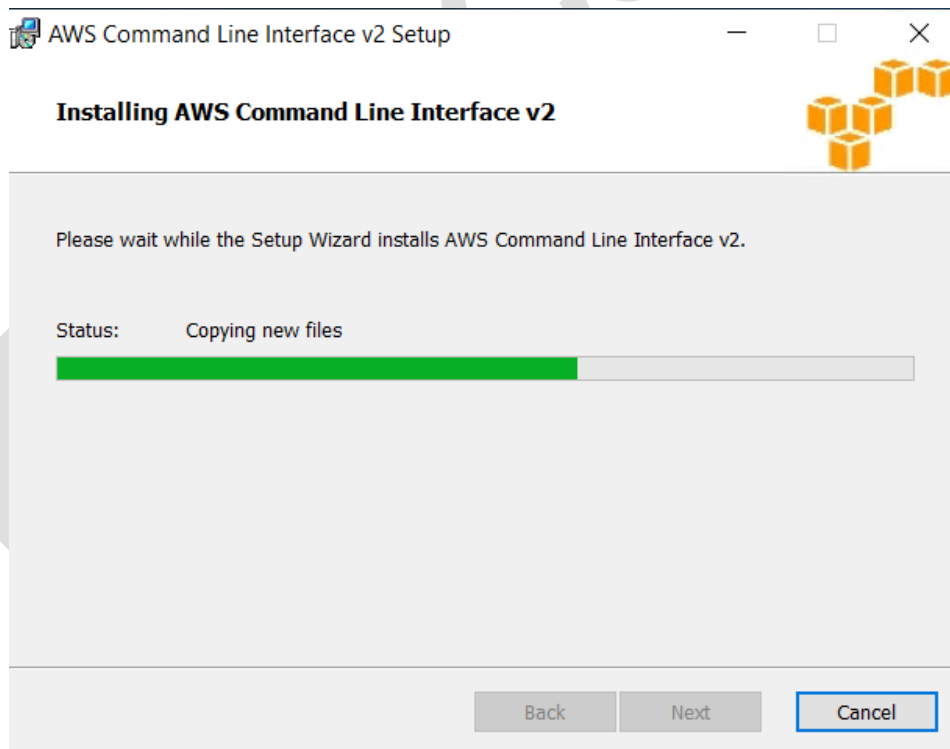
10. Keep **installation location default** and click on **Next**.



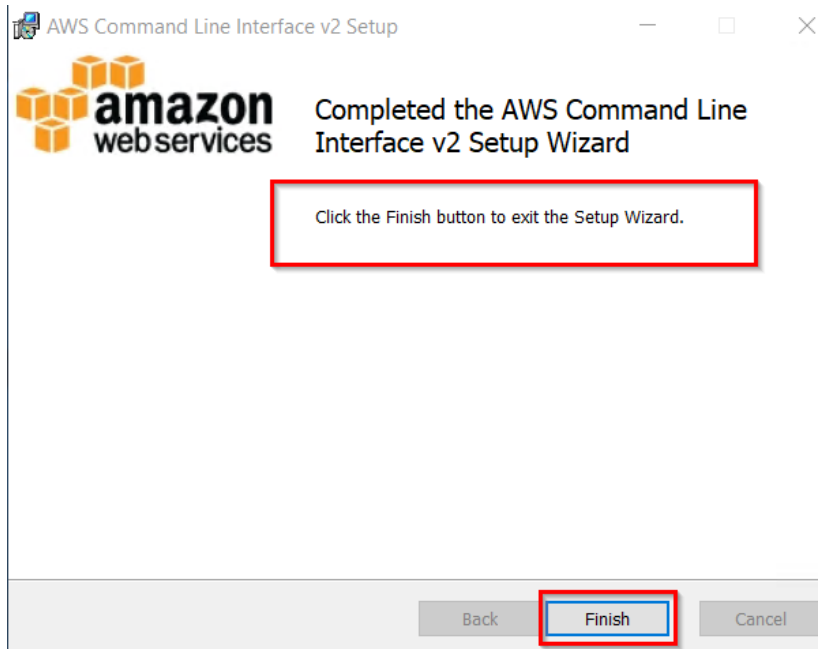
11. Click on **Install** to start installation.



12. Wait for the installation to get complete.



13. Installation has been completed click on **finish**.



14. To confirm the installation, open **command prompt (CMD)** (in your virtual machine) and use the below command.

```
aws --version
```

The screenshot shows a Windows Command Prompt window titled 'Administrator: Command Prompt'. The text inside the window reads: 'Microsoft Windows [Version 10.0.20348.1607] (c) Microsoft Corporation. All rights reserved. C:\Users\Administrator>aws --version aws-cli/2.11.9 Python/3.11.2 Windows/10 exec-env/EC2 exe/AMD64 prompt/off C:\Users\Administrator>'. The command 'aws --version' has been executed successfully, displaying the installed version and environment details.

Note: If Windows is unable to find the program and throws an error aws is not recognized as an internal or external command, then try restarting the command prompt and if the error persists then restart whole system so that the changes may take effect.

5.2 On Amazon Linux

For Amazon Linux we installed in the previous section already has CLI installed. You just need to configure CLI following in next lesson. Steps to configure CLI are same for both Windows and Linux so follow next two lessons to configure CLI on Linux.

For other Linux flavors like Ubuntu or Red Hat, follow AWS documentation at <https://docs.aws.amazon.com/cli/latest/userguide/install-cliv2-linux.html>

5.3 On macOS

We don't recommend using macOS EC2 instance, these steps are only for information purpose. We recommend you use Windows Virtual Machine.

We will be installing AWS CLI version 2.x, the current version intended for use in production environments. We will be installing AWS CLI from the terminal.

Official Documentation:

<https://docs.aws.amazon.com/cli/latest/userguide/install-cliv2-mac.html>

1. Open the terminal and run the following commands:

```
$ curl "https://s3.amazonaws.com/aws-cli/awscli-bundle.zip" -o "awscli-bundle.zip"
```

This command is using the **curl** tool to download a file from a URL and save it with a specific name.

curl: This is a command-line tool for making HTTP requests. In this case, it's being used to download a file.

"https://s3.amazonaws.com/aws-cli/awscli-bundle.zip": This is the URL of the file you want to download. The -o option is used to specify the output file, and "awscli-bundle.zip" is the name you're giving to the file once it's downloaded.

So, when you run this command, it will use curl to download the file from the specified URL and save it as "awscli-bundle.zip" in the current working directory.

```
$ unzip awscli-bundle.zip
```

This command is using the unzip tool to extract the contents of the **"awscli-bundle.zip"** file. The unzip command is commonly used for extracting files from ZIP archives.

unzip: This is the command used for extracting files from ZIP archives.

awscli-bundle.zip: This is the name of the ZIP file that you want to extract.

So, when you run this command, it will unzip the contents of "awscli-bundle.zip" in the current working directory, making the files and directories contained in the ZIP archive available for use.

```
$ sudo ./awscli-bundle/install -i /usr/local/aws -b /usr/local/bin/aws
```

This will install AWS CLI on macOS.

To confirm the install, close this terminal and open a new terminal and type the command

```
$ aws --version
```

If the output shows the version, then it is good to go.

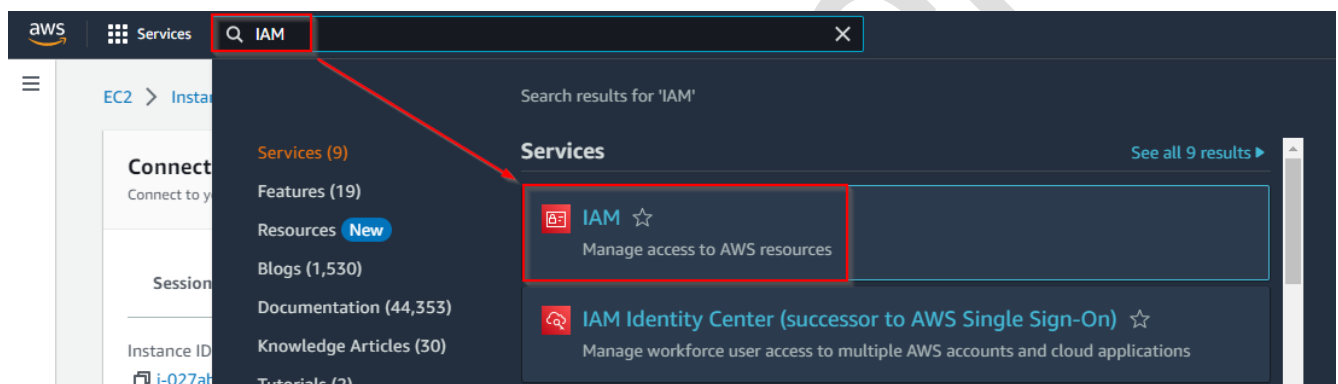
6 CREATING AWS ACCESS KEYS FOR AWS INTERACTION VIA CLI

In this section, we will be providing steps to create IAM User and download your Access key ID and your Secret access key

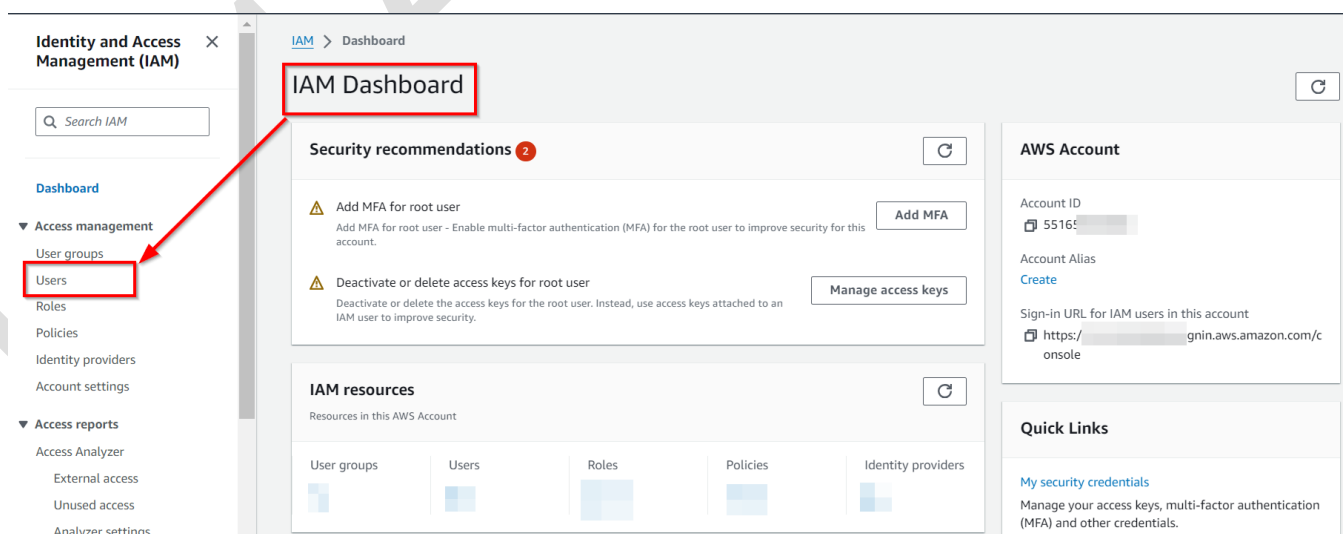
Access keys are long-term credentials for an IAM user or the AWS account root user. You can use access keys to sign **programmatic requests** to the AWS CLI or AWS API (directly or using the AWS SDK).

You must use **both** the access key ID and secret access key together to authenticate your requests. **Manage your access keys as securely as you do your username and password.**

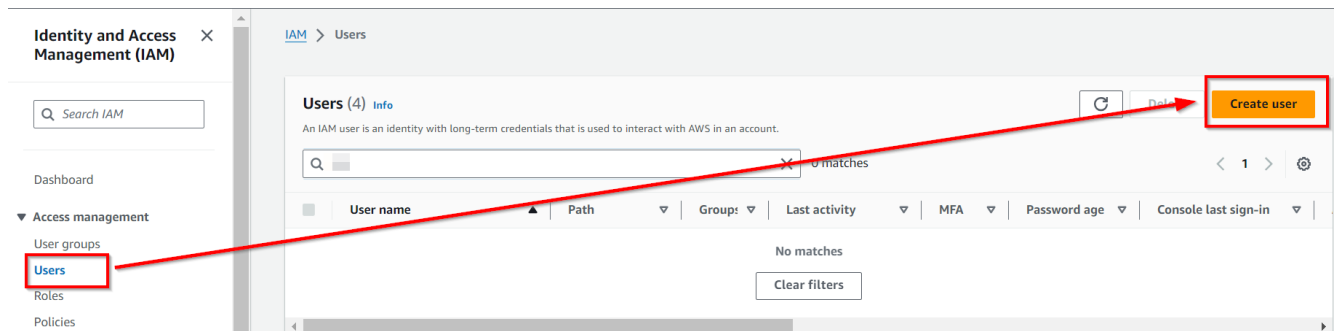
1. Log in to your AWS console and go to the IAM service section.



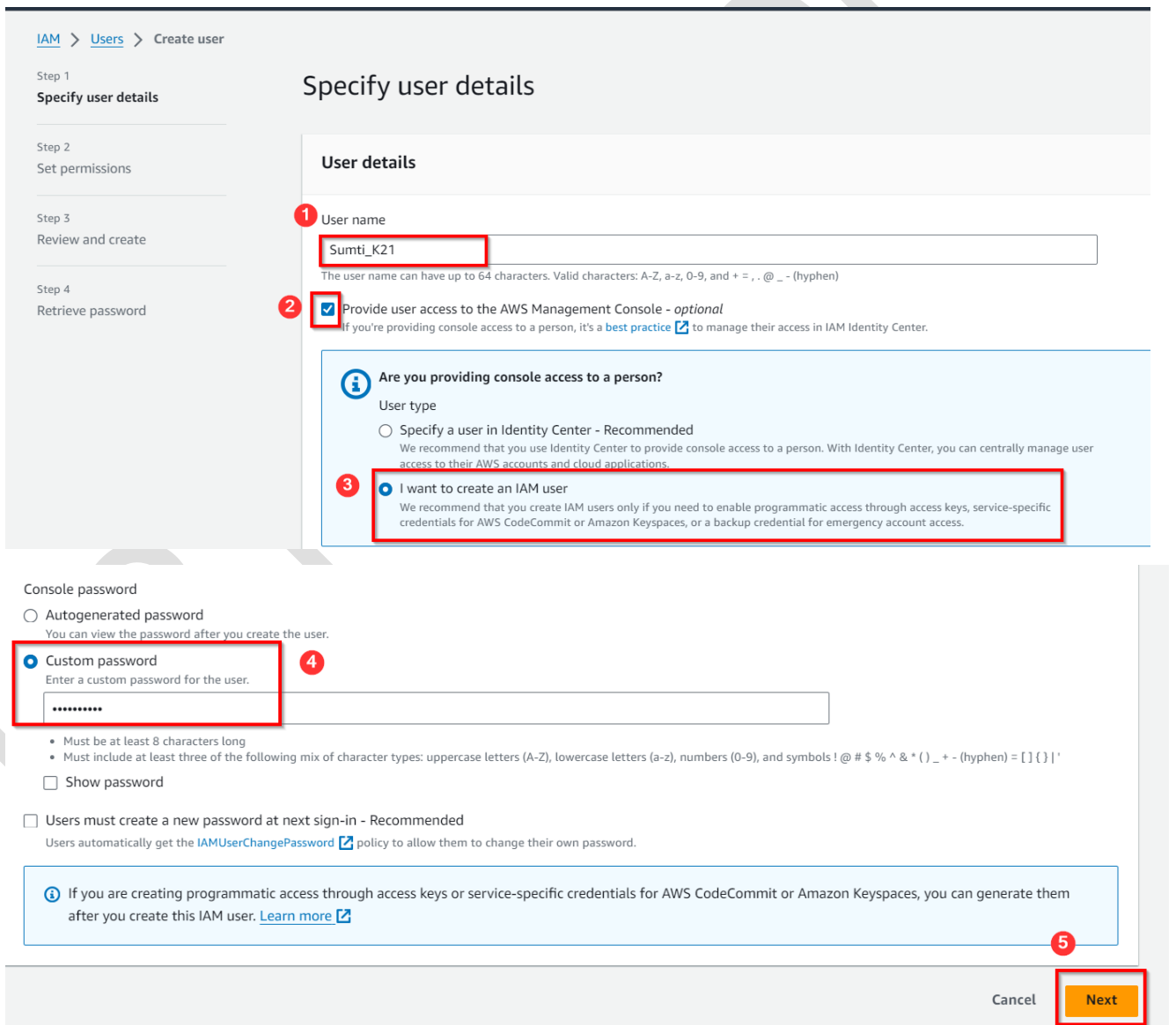
2. Under **IAM Dashboard** and then click on **Users** from left hand side navigation panel.



3. Now Click on **Add user**.



4. Provide the **username**, select Enable Console Access then to select custom password and click on **Next**.



Specify user details

User details

- User name**
Sumti_K21
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, =, ., @, _ (hyphen)
- ☒ **Provide user access to the AWS Management Console - optional**
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

Are you providing console access to a person?

User type

- ☐ **Specify a user in Identity Center - Recommended**
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.
- ☒ **I want to create an IAM user**
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☐ Autogenerated password
You can view the password after you create the user.

- ☒ **Custom password**
Enter a custom password for the user.

 - Must be at least 8 characters long
 - Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } | ' " , . / ~ `

☐ Show password

☐ Users must create a new password at next sign-in - Recommended
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

Info If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel
 Next

- Click on **Create group** as we want our user to belong to a particular group.

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☒ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1)

Search groups

< 1 > ⚙

☐	Group name	Users	Attached policies	Created
☐	k21_group	0	AdministratorAccessOnlyUser	2023-07-13 13:05:10 UTC

Create group

- Provide Group name as **Developer** and select group policy for demo we are using **Administrator access** then click on **Create Group**.

Create user group

Create a user group and select policies to attach to the group. We recommend using groups to manage user permissions by job function, AWS service access, or custom permissions. [Learn more](#)

User group name
Enter a meaningful name to identify this group.

Developers

Maximum 128 characters. Use alphanumeric and '+', '@', '-' characters.

Permissions policies (1/810)

AdministratorAccess 4 matches

☒	Policy name	Type	Used as	Description
☒	AdministratorAccess	AWS managed - job function	Permissions policy (1)	Provides full access to AWS services and re...
☐	AdministratorAccess-Amplify	AWS managed	None	Grants account administrative permissions ...
☐	AdministratorAccess-AWSElasticB...	AWS managed	None	Grants account administrative permissions...
☐	AWSAuditManagerAdministrator...	AWS managed	None	Provides administrative access to enable or...

Cancel **Create user group**

7. Now your Group is added click on **Next** button.

Permissions options

☒ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1/2)

Search groups

	Group name	Users	Attached policies	Created
☒	Developers	0	AdministratorAccess	2023-01-27 (Now)
☐	...	...	...	...

Permissions boundary - optional
Set a permissions boundary to control the maximum permissions for this user. Use this advanced feature used to delegate permission management to others. [Learn more](#)

Cancel Previous **Next**

8. Review the settings and click on **Create User**

Developers user group created.

IAM > Users > Create user

Step 1: Specify user details
Step 2: Set permissions
Step 3: Review and create
Step 4: Retrieve password

Review and create
Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name Sumit_K21	Console password type Custom password	Require password reset Yes
------------------------	--	-------------------------------

Permissions summary

Name	Type	Used as
Developers	Group	Permissions group
IAMUserChangePassword	AWS managed	Permissions policy

Tags - optional
Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

Cancel Previous **Create user**

9. Download the .csv file and click on Return to user list

User created successfully
You can view and download the user's password and email instructions for signing in to the AWS Management Console.

[View user](#)

[IAM](#) > [Users](#) > [Create user](#)

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details

Console sign-in URL
https://532661433831.signin.aws.amazon.com/console

User name
Sumti_K21

Console password
***** [Show](#)

[Email sign-in instructions](#)

[Download .csv file](#) [Return to users list](#)

10. Now go to the IAM user and click on the user you just created. You can create the Access key using two ways as snow in below.

[IAM](#) > [Users](#) > [Sumti_K21](#)

Sumti_K21 [Info](#) [Delete](#)

Summary

ARN arn:aws:iam::551659568890:user/Sumti_K21	Console access Enabled without MFA	Access key 1 Create access key
Created December 29, 2023, 14:17 (UTC+05:30)	Last console sign-in Never	

[Permissions](#) | [Groups \(1\)](#) | [Tags](#) | **[Security credentials](#)** | [Access Advisor](#)

Console sign-in

Console sign-in link
https://551659568890.signin.aws.amazon.com/console

Console password
Updated 3 minutes ago (2023-12-29 14:17 GMT+5:30)

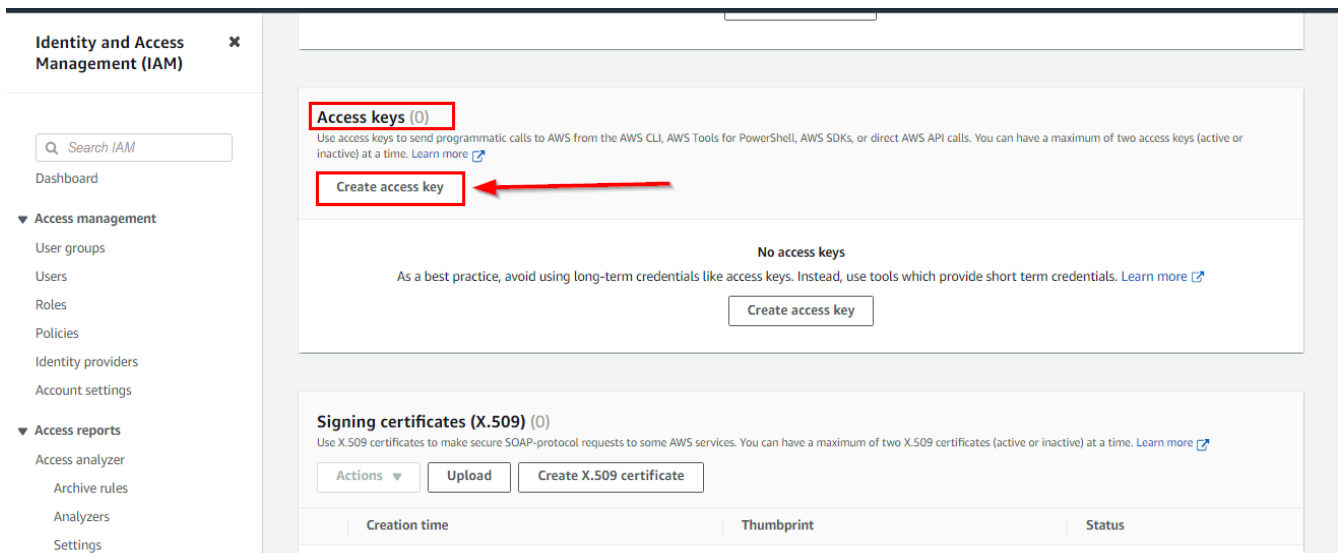
Last console sign-in
Never

[Manage console access](#)

OR
You can scroll down under this to find Access Keys

You can click on this to create the Access keys

11. Scroll down and Under the Access Keys click on Create Access Keys or you can chose Create access key option directly.



Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings

Access reports

- Access analyzer
- Archive rules
- Analyzers
- Settings

Access keys (0)

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

Create access key

No access keys

As a best practice, avoid using long-term credentials like access keys. Instead, use tools which provide short term credentials. [Learn more](#)

Create access key

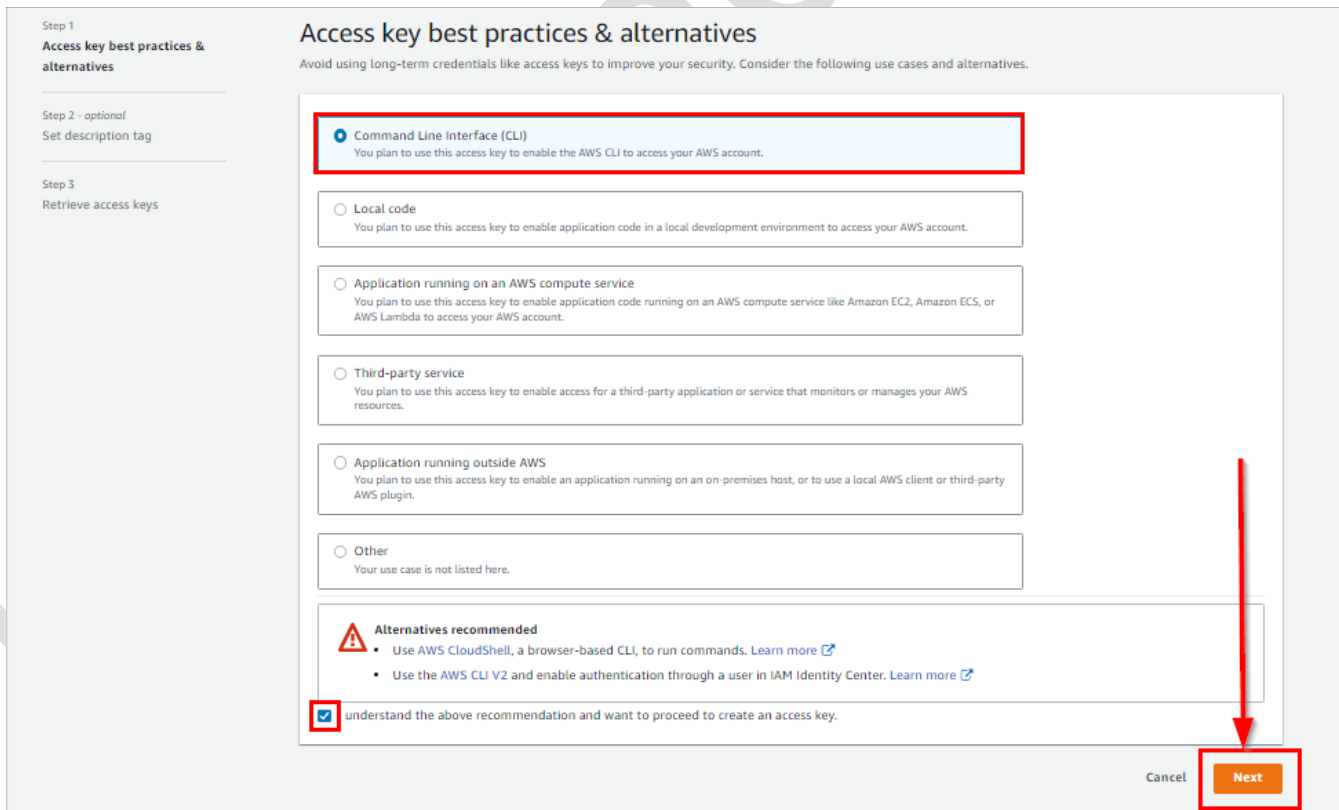
Signing certificates (X.509) (0)

Use X.509 certificates to make secure SOAP-protocol requests to some AWS services. You can have a maximum of two X.509 certificates (active or inactive) at a time. [Learn more](#)

Actions Upload Create X.509 certificate

Creation time	Thumbprint	Status
---------------	------------	--------

12. Click on **Command line interface** and select the **check box** and click on **Next**.



Step 1
Access key best practices & alternatives

Step 2 - optional
Set description tag

Step 3
Retrieve access keys

Access key best practices & alternatives

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

☒ **Command Line Interface (CLI)**
You plan to use this access key to enable the AWS CLI to access your AWS account.

☐ **Local code**
You plan to use this access key to enable application code in a local development environment to access your AWS account.

☐ **Application running on an AWS compute service**
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.

☐ **Third-party service**
You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.

☐ **Application running outside AWS**
You plan to use this access key to enable an application running on an on-premises host, or to use a local AWS client or third-party AWS plugin.

☐ **Other**
Your use case is not listed here.

Alternatives recommended

- Use **AWS CloudShell**, a browser-based CLI, to run commands. [Learn more](#)
- Use the **AWS CLI V2** and enable authentication through a user in IAM Identity Center. [Learn more](#)

☒ I understand the above recommendation and want to proceed to create an access key.

Cancel **Next**

13. Provide a Tag as key and click on **Create access key**.

IAM > Security credentials > Create access key

Step 1
Access key best practices & alternatives

Step 2 - optional
Set description tag

Step 3
Retrieve access keys

Set description tag - optional

The description for this access key will be attached to this user as a tag and shown alongside the access key.

Description tag value

Describe the purpose of this access key and where it will be used. A good description will help you rotate this access key confidently later.

key

Maximum 256 characters. Allowed characters are letters, numbers, spaces representable in UTF-8, and: _ . : / = + - @

Cancel Previous **Create access key**

14. Download the .csv file and click on **Done**.

✓ Access key created
This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.

Step 1
Access key best practices & alternatives

Step 2 - optional
Set description tag

Step 3
Retrieve access keys

Retrieve access keys [Info](#)

Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key	Secret access key
 AKIAYA4LY735OYOSWQ45	 ***** Show

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

[Download .csv file](#) **Done**

15. Now download the CSV file. This file contains user details like Access key id & Secret key.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Access key ID	Secret access key										
2	AKIAXYBIYOXT7BN	mDndO0wuSWYB5lY679bmJr3u2nvCgEXt9+vjrO4o										
3												
4												
5												
6												
7												
8												
9												

Note: After you successfully create your IAM User, you will get **one chance** to download your **Access key ID** and your **Secret access key**. These are essential for us to be able to log in to the AWS CLI, so click on the Download .csv file to download those keys and store them somewhere safe.

Note: Do **not** provide your access keys to a third party. By doing this, you might give someone permanent access to your account.

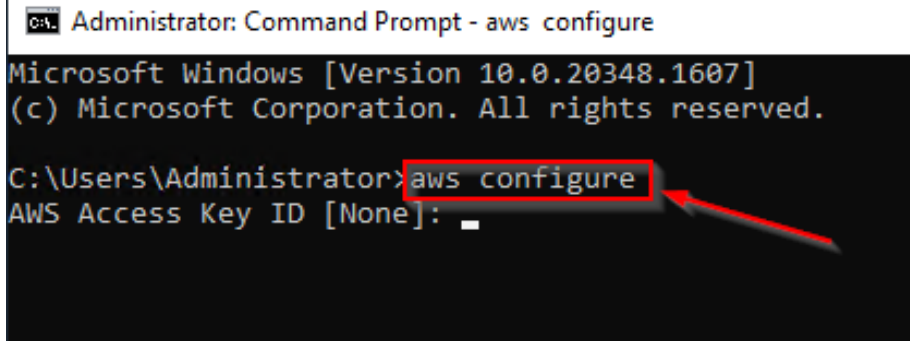
7 CONFIGURING AWS CLI TO ACCESS AWS SERVICES

This section explains how to configure the settings that the AWS Command Line Interface (AWS CLI) uses to interact with AWS. These include your security credentials, the default output format, and the default AWS Region.

Note: AWS requires that all incoming requests are cryptographically signed. The AWS CLI does this for you. The "signature" includes a date/time stamp. Therefore, you must ensure that your computer's date and time are set correctly. If you don't, and the date/time in the signature is too far off of the date/time recognized by the AWS service, AWS rejects the request.

1. Open a **command prompt** on your Windows Virtual machine
2. To configure the installation, run following command

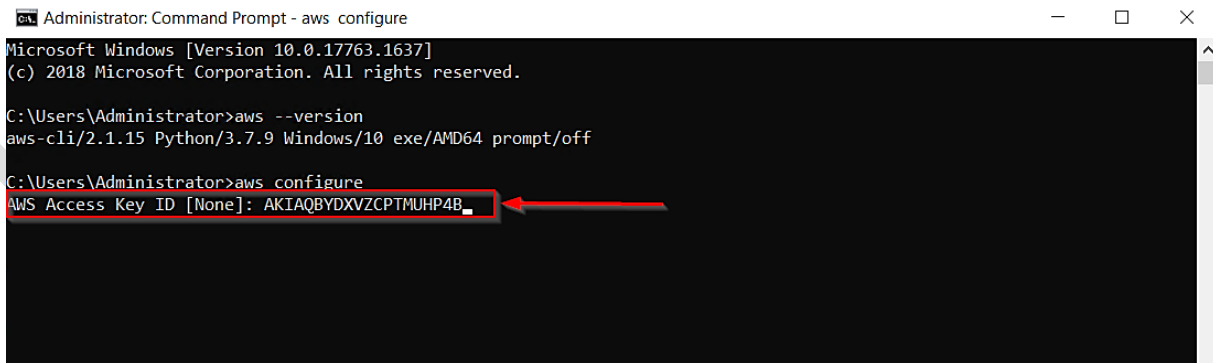
```
aws configure
```



```
Administrator: Command Prompt - aws configure  
Microsoft Windows [Version 10.0.20348.1607]  
(c) Microsoft Corporation. All rights reserved.  
C:\Users\Administrator>aws configure  
AWS Access Key ID [None]:
```

Note: By default, all the values are set to null input except default output format.

3. Provide **AWS Access Key ID** from the downloaded **CSV file**.



```
Administrator: Command Prompt - aws configure  
Microsoft Windows [Version 10.0.17763.1637]  
(c) 2018 Microsoft Corporation. All rights reserved.  
C:\Users\Administrator>aws --version  
aws-cli/2.1.15 Python/3.7.9 Windows/10 exe/AMD64 prompt/off  
C:\Users\Administrator>aws configure  
AWS Access Key ID [None]: AKIAQBYDXVZCPTMUHP4B
```

4. Provide **AWS secret access key** from the downloaded CSV file.

```
Administrator: Command Prompt - aws configure
Microsoft Windows [Version 10.0.17763.1637]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>aws --version
aws-cli/2.1.15 Python/3.7.9 Windows/10 exe/AMD64 prompt/off

C:\Users\Administrator>aws configure
AWS Access Key ID [None]: AKIAQBYDXVZCPTMUHP4B
AWS Secret Access Key [None]: 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_
```

Note: these are keys specifically associated with an IAM or Root User, they are used to sign programmatic requests that you make to AWS via the CLI.

5. Provide **us-east-1** as **Default region name**

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.1607]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>aws configure
AWS Access Key ID [None]: AKIARXDJLHU3BX6VNYNL
AWS Secret Access Key [None]: Mmw1xUolYVVKpZ8ubfURZYmh31RBdNL5TegkX1PY
Default region name [None]: us-east-1
```

Note: For example, I have typed **us-east-1** to use US East (Virginia).

Note: This identifies the AWS Region whose servers you want to send your requests to by default. This is typically the Region closest to you, but it can be any Region.

6. Select **json** as **Default output format**.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.1607]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>aws configure
AWS Access Key ID [None]: AKIARXDJLHU3BX6VNYNL
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json

C:\Users\Administrator>
```

Note: This specifies how the results are formatted. The value can be any of the values in the following list. If you don't specify an output format, json is used as the default.

Note: All the key details will remain persistent even after closing cmd prompt.

Note: You have 4 choices for the default output format.

1. **json** – The output is formatted as a JSON string.
2. **yaml** – The output is formatted as a YAML string. (Available in the AWS CLI version 2 only.)
3. **text** – The output is formatted as multiple lines of tab-separated string values. This can be useful to pass the output to a text processor, like grep, sed, or awk.
4. **table** – The output is formatted as a table using the characters +|- to form the cell borders. It typically presents the information in a "human-friendly" format that is much easier to read than the others, but not as programmatically useful.
7. It will take a few minutes and then drop you back into command prompt.

Note: With this your AWS CLI access is setup and ready to go, you can enter a myriad of commands to communicate with each and every AWS service from your command line now. In further steps let's try some CLI commands to access S3 service.

7.1 Create and Delete S3 Bucket Using AWS CLI

Note: Till now, we have successfully installed AWS CLI in our Windows Virtual Machine and login with our access keys. Here in further steps, we shall try to use CLI commands to access AWS Services. Now, we shall create a S3 bucket using AWS CLI and check on the S3 console whether it is created successfully or not, and then once verified we shall delete the bucket using AWS CLI commands, and validate on the S3 console it shall reflect for the same.

1. Once you have successfully installed AWS CLI, run the following commands to create a **S3 bucket using CLI**.

```
aws s3 mb s3://yourbucketname  
aws s3 ls
```

Note: An Amazon S3 bucket name should be globally unique as the namespace is shared by all AWS accounts. This means that after a bucket is created, the name of that bucket cannot be used by another AWS account in any AWS Region until the bucket is deleted. The following rules apply for naming S3 buckets. So, make sure you follow all the rules before naming the S3 bucket:

- a. **Bucket names must be between 3 and 63 characters long.**
- b. **Bucket names can consist only of lowercase letters, numbers, dot (.), and hyphens (-).**
- c. **Bucket names must begin and end with a letter or number.**

```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.1607]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>aws configure
AWS Access Key ID [None]: AKIARXD7LHU3BX6VNYNL
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json

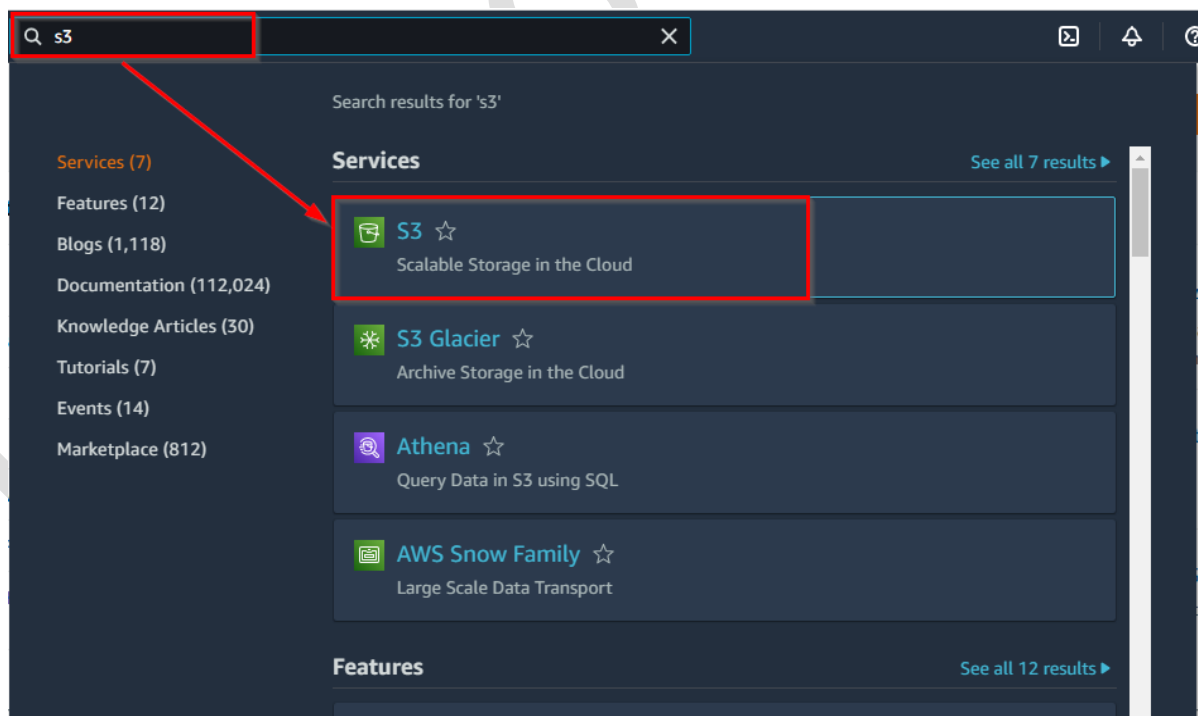
C:\Users\Administrator>aws s3 mb s3://k21bucket11
make_bucket: k21bucket11

C:\Users\Administrator>aws s3 ls
2022-11-26 15:15:28 elasticbeanstalk-us-east-1-118332538166
2023-04-05 11:15:32 k21bucket11
2023-03-04 12:09:44 my-bucket-testap
2023-03-04 11:57:39 my-state-file
2023-03-04 16:38:43 my-terraform-bucket-xxxxx
2023-04-04 07:18:46 node-sdk-sample-aba4fb22-ae53-4d06-a361-a544a1612fd5
2023-03-04 15:35:57 terraform-day5-state

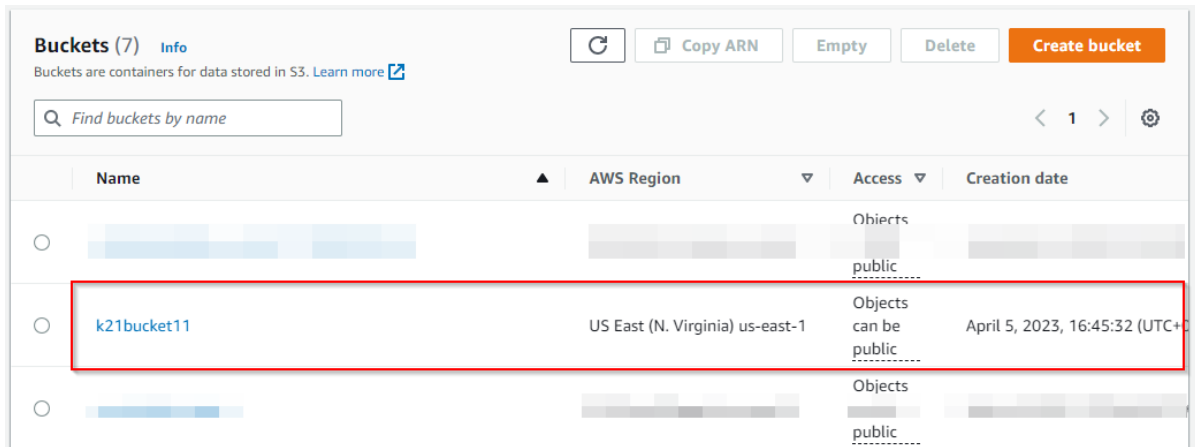
C:\Users\Administrator>_
  
```

Note: 'aws s3 ls' command shall list all the buckets present in your account. As this will be your **first S3 bucket** you shall see only a **single bucket** in the output after running 'aws s3 ls' command.

- Now we have created a s3 bucket successfully using CLI, let's **verify** on **S3 console** whether it is present there or not. For that **search for S3** on your AWS console and click on it to **open**.



3. Here you shall see that the bucket we created earlier is **present**.



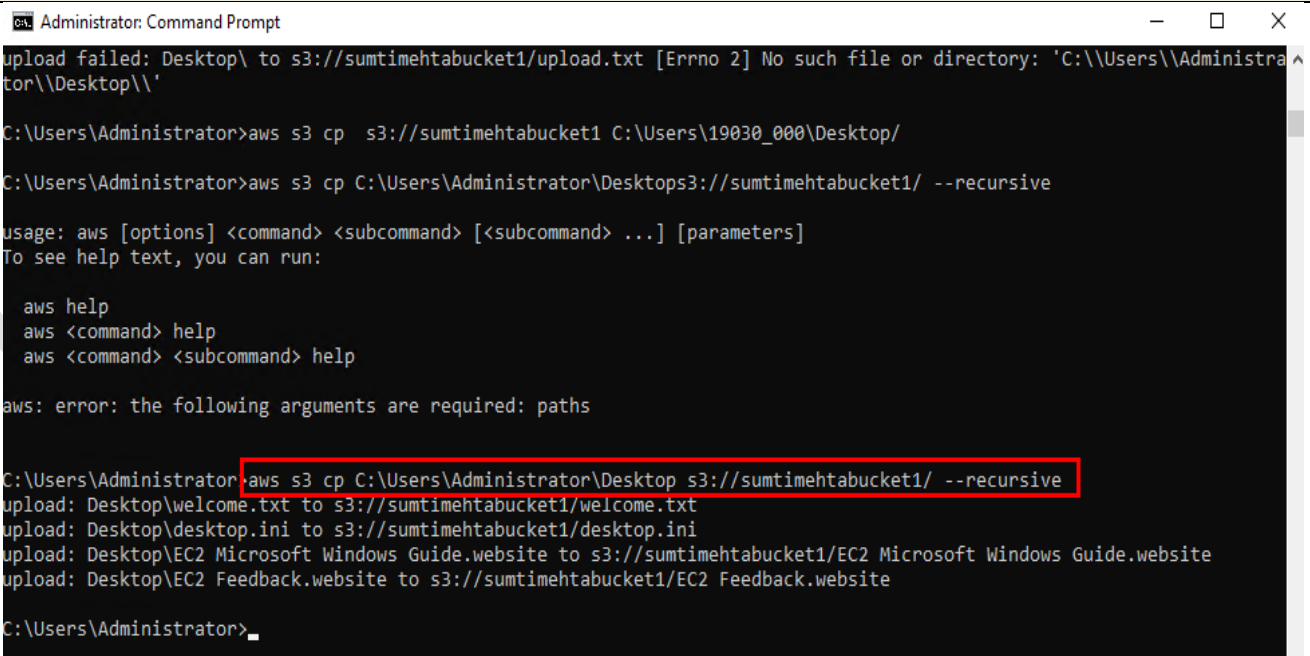
Note: As this shall be your **first bucket**, you will see only **one bucket present** here on the **S3 console**.

4. To upload objects to the bucket you need to specify the source address of the object and destination address of the bucket.

we are uploading it via the virtual machine, so you need to copy the file to virtual machine and provide the source address within the virtual machine.

```
aws s3 cp C:\source\address s3://yourbucketname / --recursive
```

Here **C:\Users\Administrator\Desktop** is source address of object
and **s3://yourbucketname /** is destination address of bucket



```
Administrator: Command Prompt
upload failed: Desktop\ to s3://sumtimehtabucket1/upload.txt [Errno 2] No such file or directory: 'C:\Users\Administrator\Desktop\'

C:\Users\Administrator>aws s3 cp s3://sumtimehtabucket1 C:\Users\19030_000\Desktop/

C:\Users\Administrator>aws s3 cp C:\Users\Administrator\Desktops3://sumtimehtabucket1/ --recursive

usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:

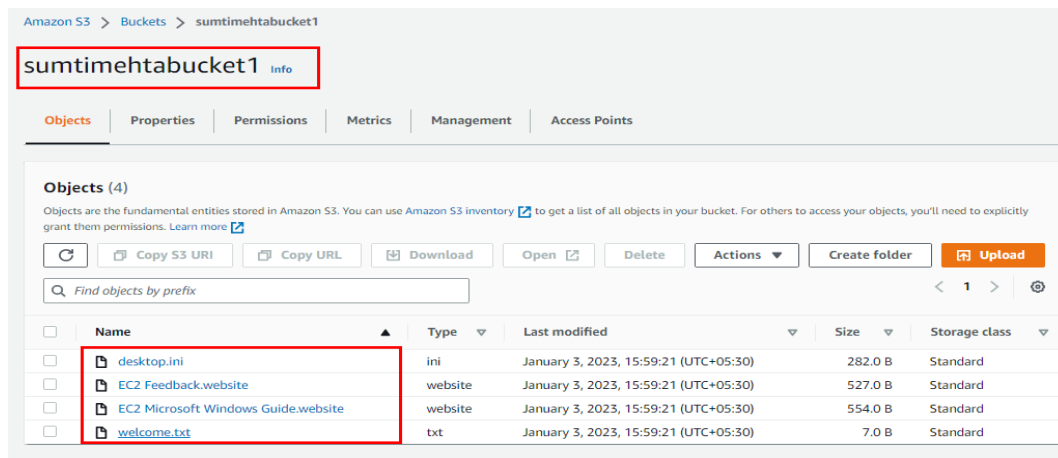
    aws help
    aws <command> help
    aws <command> <subcommand> help

aws: error: the following arguments are required: paths

C:\Users\Administrator>aws s3 cp C:\Users\Administrator\Desktop s3://sumtimehtabucket1/ --recursive
upload: Desktop\welcome.txt to s3://sumtimehtabucket1/welcome.txt
upload: Desktop\desktop.ini to s3://sumtimehtabucket1/desktop.ini
upload: Desktop\EC2 Microsoft Windows Guide.website to s3://sumtimehtabucket1/EC2 Microsoft Windows Guide.website
upload: Desktop\EC2 Feedback.website to s3://sumtimehtabucket1/EC2 Feedback.website

C:\Users\Administrator>
```

5. We can check it Manually by going to the S3 buckets in our AWS Console.



We have successfully uploaded the Objects in the bucket using CLI.

6. Now let's delete the bucket using CLI, so back on your CLI run the following commands.

```
aws s3 rb s3://yourbucketname
```

```
Administrator: Command Prompt

C:\Users\Administrator>aws s3 cp C:\Users\Administrator\Desktop\3://sumtimehtabucket1/ --recursive
usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:

    aws help
    aws <command> help
    aws <command> <subcommand> help

aws: error: the following arguments are required: paths

C:\Users\Administrator>aws s3 cp C:\Users\Administrator\Desktop s3://sumtimehtabucket1/ --recursive
upload: Desktop\welcome.txt to s3://sumtimehtabucket1/welcome.txt
upload: Desktop\desktop.ini to s3://sumtimehtabucket1/desktop.ini
upload: Desktop\EC2 Microsoft Windows Guide.website to s3://sumtimehtabucket1/EC2 Microsoft Windows Guide.website
upload: Desktop\EC2 Feedback.website to s3://sumtimehtabucket1/EC2 Feedback.website

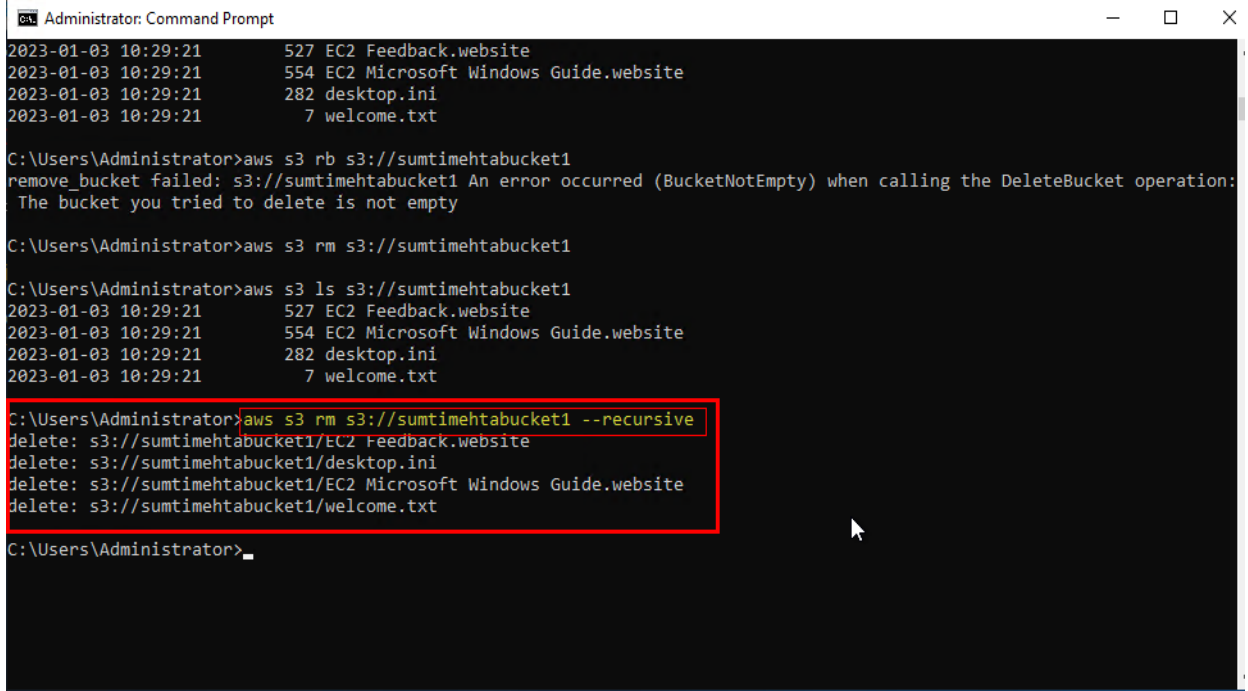
C:\Users\Administrator>aws s3 ls s3://sumtimehtabucket1/
2023-01-03 10:29:21      527 EC2 Feedback.website
2023-01-03 10:29:21      554 EC2 Microsoft Windows Guide.website
2023-01-03 10:29:21      282 desktop.ini
2023-01-03 10:29:21         7 welcome.txt

C:\Users\Administrator>aws s3 rb s3://sumtimehtabucket1
remove bucket failed: s3://sumtimehtabucket1 An error occurred (BucketNotEmpty) when calling the DeleteBucket operation:
The bucket you tried to delete is not empty

C:\Users\Administrator>
```

- Try to remove the objects present in the bucket first

```
aws s3 rm s3://yourbucketname --recursive
```



```
Administrator: Command Prompt
2023-01-03 10:29:21      527 EC2 Feedback.website
2023-01-03 10:29:21      554 EC2 Microsoft Windows Guide.website
2023-01-03 10:29:21      282 desktop.ini
2023-01-03 10:29:21         7 welcome.txt

C:\Users\Administrator>aws s3 rb s3://sumtimehtabucket1
remove_bucket failed: s3://sumtimehtabucket1 An error occurred (BucketNotEmpty) when calling the DeleteBucket operation:
The bucket you tried to delete is not empty

C:\Users\Administrator>aws s3 rm s3://sumtimehtabucket1

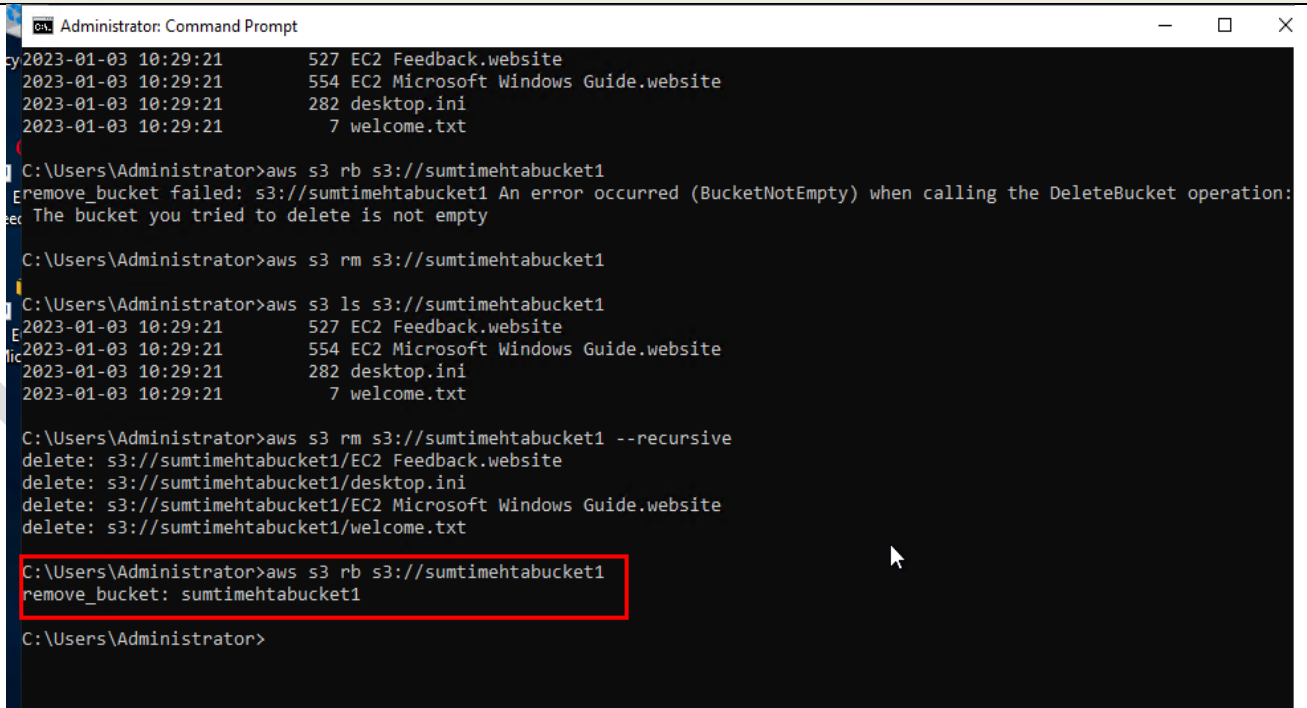
C:\Users\Administrator>aws s3 ls s3://sumtimehtabucket1
2023-01-03 10:29:21      527 EC2 Feedback.website
2023-01-03 10:29:21      554 EC2 Microsoft Windows Guide.website
2023-01-03 10:29:21      282 desktop.ini
2023-01-03 10:29:21         7 welcome.txt

C:\Users\Administrator>aws s3 rm s3://sumtimehtabucket1 --recursive
delete: s3://sumtimehtabucket1/EC2 Feedback.website
delete: s3://sumtimehtabucket1/desktop.ini
delete: s3://sumtimehtabucket1/EC2 Microsoft Windows Guide.website
delete: s3://sumtimehtabucket1/welcome.txt

C:\Users\Administrator>
```

- Now again try to delete the bucket using below command.

```
aws s3 rb s3://yourbucketname
```



```
Administrator: Command Prompt
2023-01-03 10:29:21      527 EC2 Feedback.website
2023-01-03 10:29:21      554 EC2 Microsoft Windows Guide.website
2023-01-03 10:29:21      282 desktop.ini
2023-01-03 10:29:21         7 welcome.txt

C:\Users\Administrator>aws s3 rb s3://sumtimehtabucket1
remove_bucket failed: s3://sumtimehtabucket1 An error occurred (BucketNotEmpty) when calling the DeleteBucket operation:
The bucket you tried to delete is not empty

C:\Users\Administrator>aws s3 rm s3://sumtimehtabucket1

C:\Users\Administrator>aws s3 ls s3://sumtimehtabucket1
2023-01-03 10:29:21      527 EC2 Feedback.website
2023-01-03 10:29:21      554 EC2 Microsoft Windows Guide.website
2023-01-03 10:29:21      282 desktop.ini
2023-01-03 10:29:21         7 welcome.txt

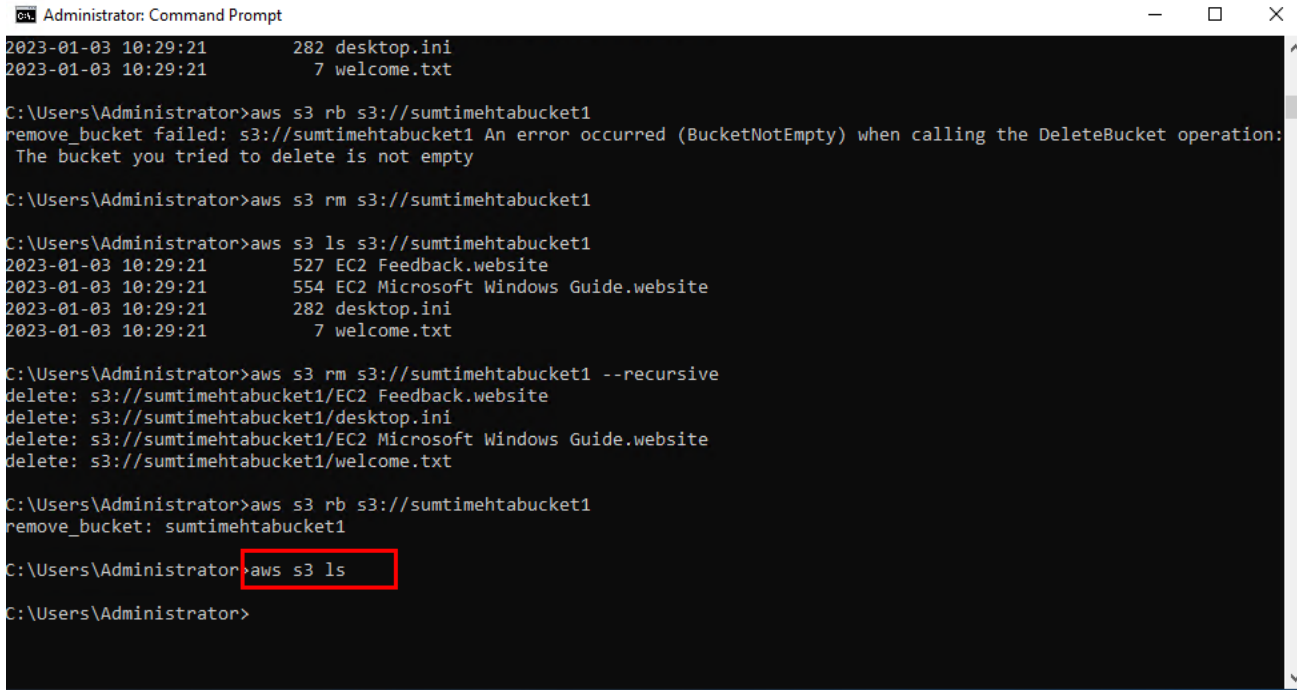
C:\Users\Administrator>aws s3 rm s3://sumtimehtabucket1 --recursive
delete: s3://sumtimehtabucket1/EC2 Feedback.website
delete: s3://sumtimehtabucket1/desktop.ini
delete: s3://sumtimehtabucket1/EC2 Microsoft Windows Guide.website
delete: s3://sumtimehtabucket1/welcome.txt

C:\Users\Administrator>aws s3 rb s3://sumtimehtabucket1
remove_bucket: sumtimehtabucket1

C:\Users\Administrator>
```

9. Now check if your bucket is deleted or not.

```
aws s3 ls
```



```
Administrator: Command Prompt
2023-01-03 10:29:21      282 desktop.ini
2023-01-03 10:29:21         7 welcome.txt

C:\Users\Administrator>aws s3 rb s3://sumtimehtabucket1
remove_bucket failed: s3://sumtimehtabucket1 An error occurred (BucketNotEmpty) when calling the DeleteBucket operation:
The bucket you tried to delete is not empty

C:\Users\Administrator>aws s3 rm s3://sumtimehtabucket1

C:\Users\Administrator>aws s3 ls s3://sumtimehtabucket1
2023-01-03 10:29:21      527 EC2 Feedback.website
2023-01-03 10:29:21      554 EC2 Microsoft Windows Guide.website
2023-01-03 10:29:21      282 desktop.ini
2023-01-03 10:29:21         7 welcome.txt

C:\Users\Administrator>aws s3 rm s3://sumtimehtabucket1 --recursive
delete: s3://sumtimehtabucket1/EC2 Feedback.website
delete: s3://sumtimehtabucket1/desktop.ini
delete: s3://sumtimehtabucket1/EC2 Microsoft Windows Guide.website
delete: s3://sumtimehtabucket1/welcome.txt

C:\Users\Administrator>aws s3 rb s3://sumtimehtabucket1
remove_bucket: sumtimehtabucket1

C:\Users\Administrator>aws s3 ls

C:\Users\Administrator>
```

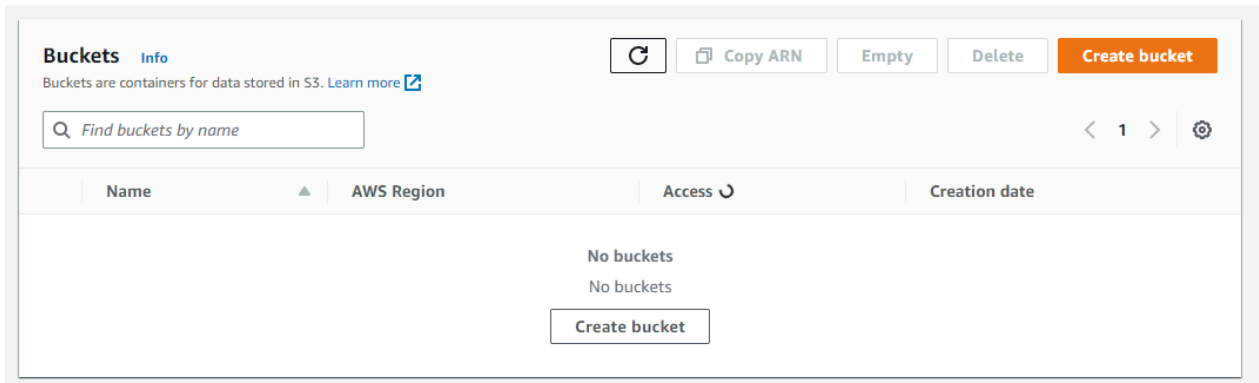
- ls stands for List
- rb stands for Remove Bucket
- rm is used to remove the object from the bucket.

Note: You shall see that the bucket is not present now while listing by 'aws s3 ls' command. You can now validate whether it has been deleted from the S3 console or not. Back on console please refresh and then you shall see that the bucket has been disappeared from there i.e. we were able to successfully delete the bucket using CLI.

You can empty a bucket using the AWS CLI only if the bucket does not have Bucket Versioning enabled. If versioning is not enabled, you can use the rm (remove) AWS CLI command with the --recursive parameter to empty the bucket (or remove a subset of objects with a specific key name prefix)

Note: In this lab, we are just giving an overview of how AWS CLI works. We won't use the CLI often. We will cover IAM users and S3 in more detail in upcoming labs.

10. Back on your S3 console, **refresh** and you shall see the bucket has been **disappeared**.



Thus, we were able to successfully create and delete the s3 bucket using AWS CLI.

7.2 Describe EC2 instance via CLI

Note: This is an optional command through which you can check the configuration of your running instance on your AWS Console through the CLI by running the following Command. To get this command executed, an instance should be in a running state and you must be inside your AWS Environment as discussed in [Topic7](#)

```
aws ec2 describe-instances
```

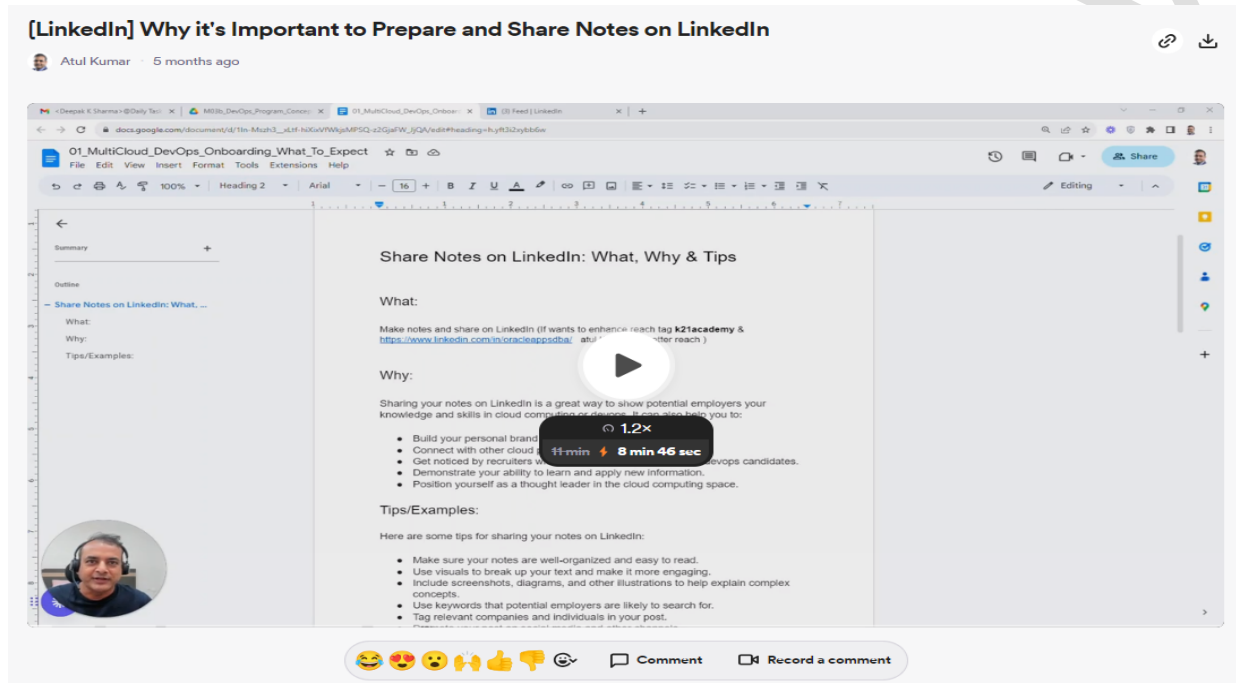
```
C:\Users\Administrator>aws ec2 describe-instances
{
  "Reservations": [
    {
      "Groups": [],
      "Instances": [
        {
          "AmiLaunchIndex": 0,
          "ImageId": "ami-0e38fa17744b2f6a5",
          "InstanceId": "i-027ab584a27b51435",
          "InstanceType": "t2.micro",
          "KeyName": "Windows-key",
          "LaunchTime": "2023-04-05T10:45:25+00:00",
          "Monitoring": {
            "State": "disabled"
          },
          "Placement": {
            "AvailabilityZone": "us-east-1c",
            "GroupName": "",
            "Tenancy": "default"
          },
          "Platform": "windows",
          "PrivateDnsName": "ip-172-31-94-81.ec2.internal",
          "PrivateIpAddress": "172.31.94.81",
          "ProductCodes": [],
          "PublicDnsName": "ec2-3-87-56-67.compute-1.amazonaws.com",
          "PublicIpAddress": "3.87.56.67",
          "State": {
            "Code": 16,
            "Name": "running"
          }
        }
      ]
    }
  ]
}
```

8 SHARE YOUR LEARNINGS

In this section, you are going to share whatever you learned during this lab on LinkedIn and WhatsApp.

Please watch the below video to understand how you can achieve this:

<https://www.loom.com/share/d91f9abbec794f4f85e7d99bb82c2ff6?sid=5ecd766a-6968-4c44-ac42-9c0ece8928d5>



🧑 As you know, building a strong professional profile is essential in today's job market. One way to showcase your skills and knowledge is by sharing your labs and projects on LinkedIn.

Your LinkedIn profile is a powerful tool in your job search. Sharing your labs and projects is just one way to demonstrate your expertise and stand out to potential employers. If you don't have LinkedIn, we strongly recommend you create one for yourself.

📝 By doing so, you can demonstrate to potential employers or connections that you have hands-on experience in your field and are actively engaged in learning and growing your skills.

8.1 On LinkedIn

Take a screenshot of lab and share it on your LinkedIn. So, this will attract recruiters and employers on your profile and increase reach. Do remember to tag **K21Academy**, as we will circulate in our network too, to increase your reach.

Please refer these following examples on how to share on LinkedIn (don't limit yourself to just these type). Share learnings in points, notes 📝 or anything that can help others.

Here is a sample which includes steps inside a post followed by screenshot:

Sumti Mehta • 1st
Cloud Computing | Amazon Web Services | 1x AWS Certified | Technical Support...
2w • Edited • 🌐

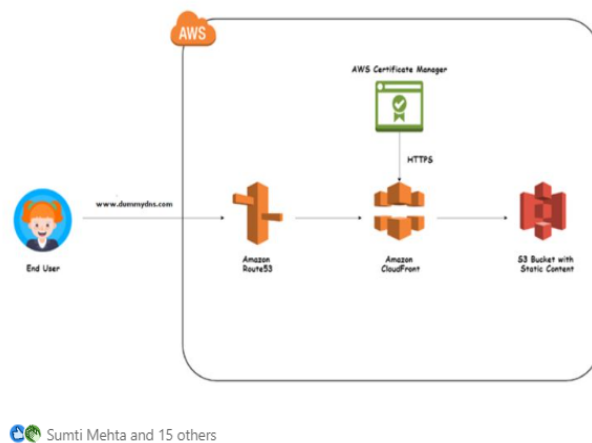
Hi There,

Today I have Provisioned an SSL/TLS Certificates using AWS Certificate Manager. It was fun and learned a lot of services integrated in the deployment.

To do this deployment I had to gone through the following steps:

1. Created an S3 bucket and enabled Static Website Hosting
2. Created a Hosted Zone and Mapped DNS Using Route53
3. Requested SSL/TLS certificates using AWS Certificate Manager
4. And finally, Configured CloudFront Distribution

Thanks, [K21Academy: Learn Cloud From Experts](#) For providing me with the required resources. [#aws](#) [#awscloud](#) [#awsservices](#) [#learning](#) [#cloud](#) [#cloud](#) [#project](#) [#learning](#)



Sahid Anwar Hussain • 1st
DevOps Engineer & Cloud Support at K21Academy: Learn Cloud From Experts
2w • 🌐

Hi There,

Today I have deployed an awesome tool to monitor an entire K8s cluster. It was fun and learned a lot diring the deployment.

To do this deployment I had to gone through the following steps:

1. Created an multinode K8s cluster.
2. Created a namespace named "monitoring".
3. Created clusterRole & clusterRole Binding for Prometheus.
4. Created the ConfigMap enclosing all the necessary Data for deploying Prometheus
5. Created a Deployment file for Prometheus
6. Finally I ran all the files and wallah.. Its done.

Thanks, [K21Academy: Learn Cloud From Experts](#) For providing me with the required resources. [#kubernetes](#) [#k8s](#) [#devops](#) [#learning](#) [#cloud](#) [#cloud](#) [#project](#) [#learning](#) [#prometheus](#) [#monitoring](#)

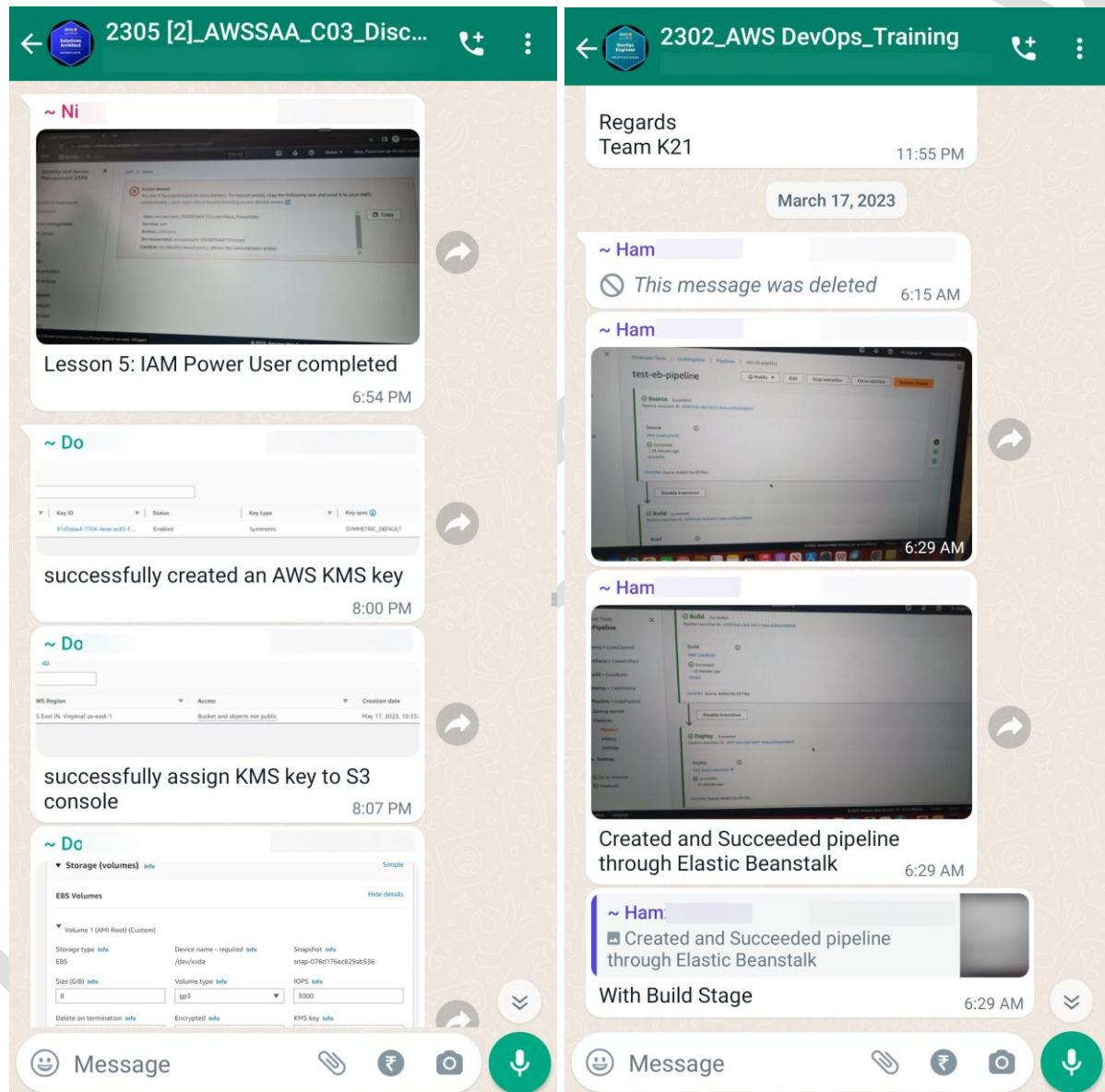
The screenshot shows the Prometheus web interface with the 'Targets' tab selected. It displays a table of scraped targets with columns for Endpoint, State, Labels, Last Scrape, and Scrape Duration. The targets include kube-state-metrics, kubernetes-apiservers, and kubernetes-cadvisor.

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://kubernetes-apiservers:8080/metrics	UP	kubernetes-apiservers	1.99s ago	12.42ms	
http://kubernetes-cadvisor:8080/metrics	UP	kubernetes-cadvisor	1.99s ago	95.47ms	

8.2 On WhatsApp

Share screenshot of lab in WhatsApp group so this will encourage others and also boost your confidence on the progressive path you are following.

Also, we will keep track of your progress on labs so do share your screenshot in WhatsApp. group too.



9 CLEAN- UP

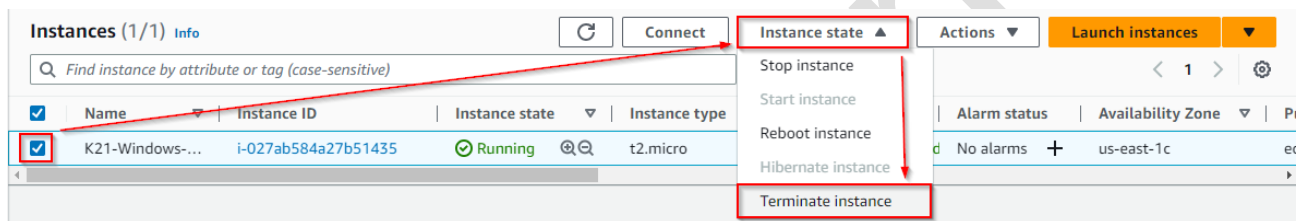
In this section we can keep the buckets and user because those are not chargeable.

Here we configure everything on Windows instance, so we only need to stop or terminate the windows instance.

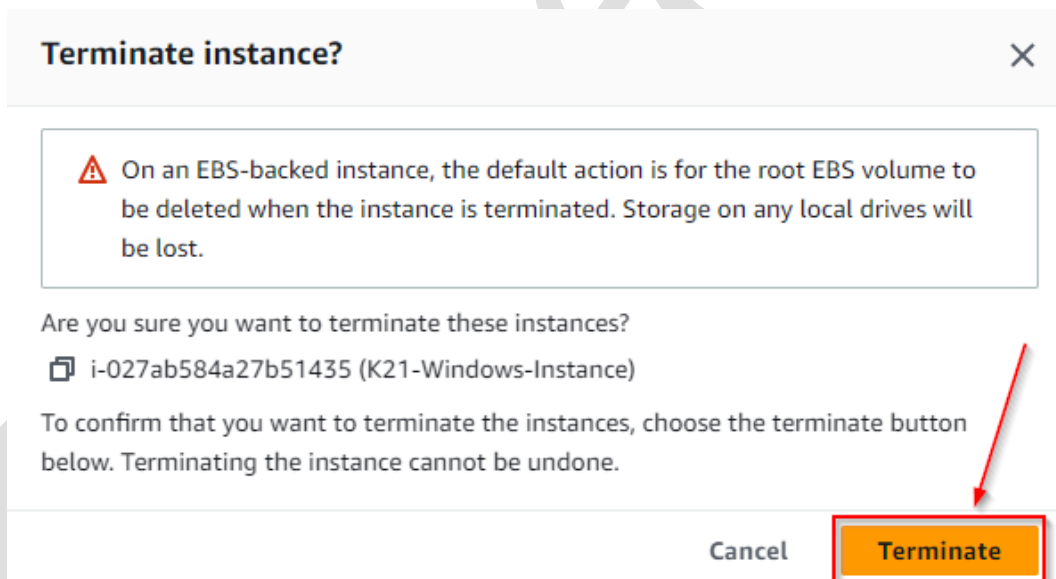
9.1 Terminating the Windows Instance

Note: This section will cover steps to terminate the windows instance

1. Go to your Instance and select the Windows Instance. Click on **Instance State** and click on **Terminate**



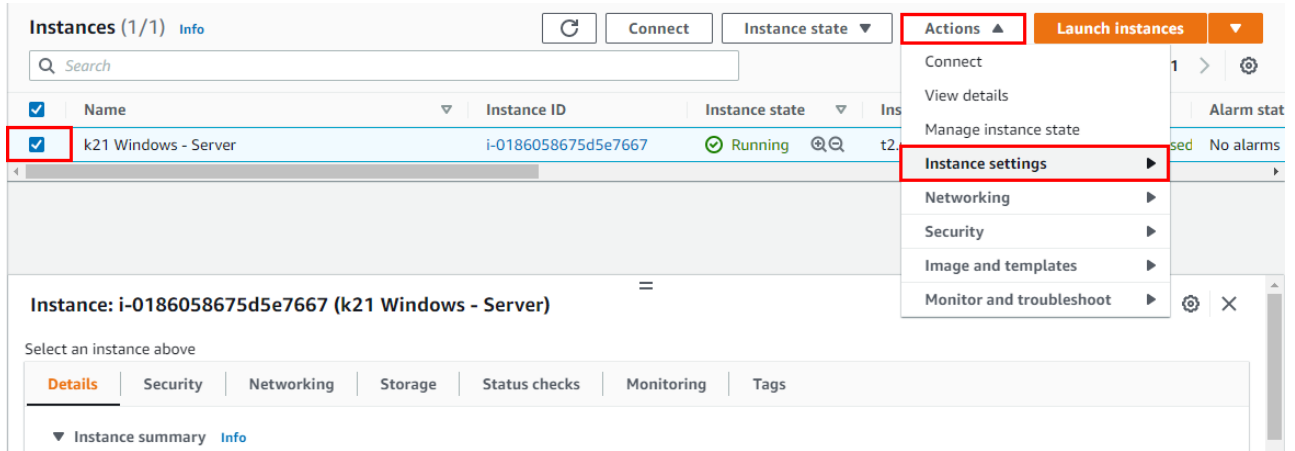
2. Click on **Terminate** to terminate the Instance



9.2 Termination Prevention of Instance

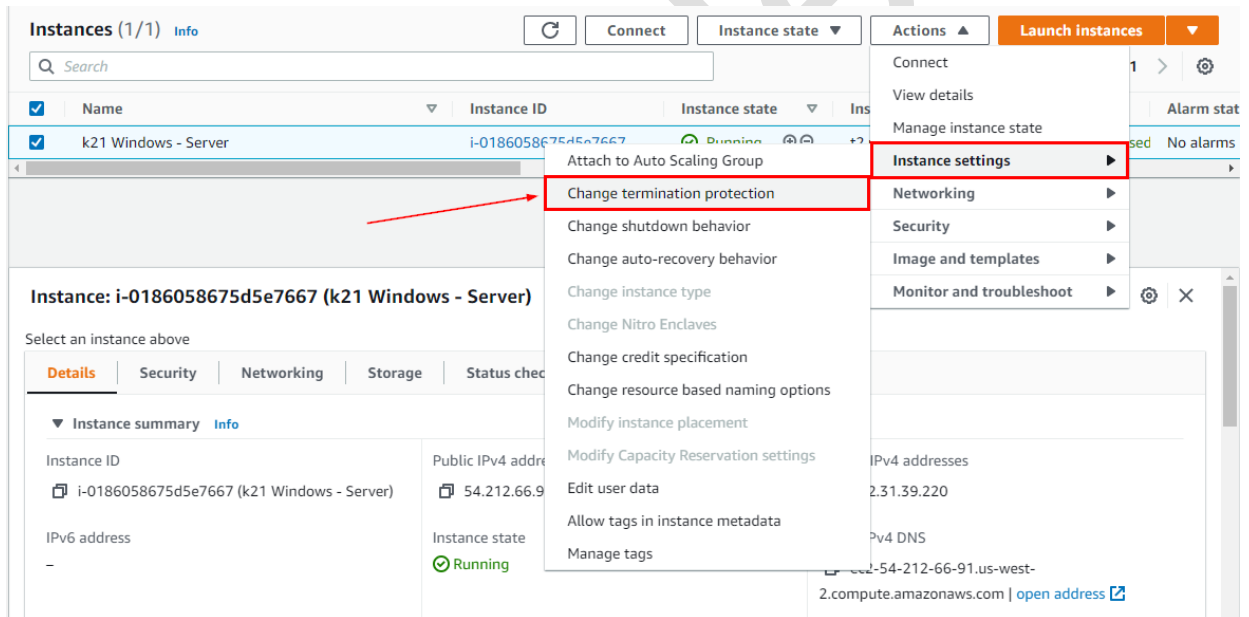
Note: To protect against the data loss caused by accidental termination of an Amazon EC2 instance, you should enable the termination protection and follow these steps to enable/disable Termination Protection.

1. Select the **Instance**, click on **Action** and then click on the **Instance settings**



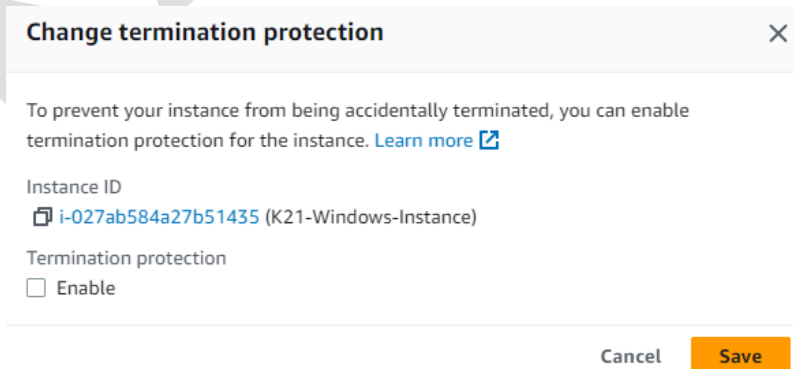
The screenshot shows the AWS Management Console 'Instances' page. A table lists instances, with 'k21 Windows - Server' (ID: i-0186058675d5e7667) in a 'Running' state. The 'Actions' dropdown menu is open, showing options like 'Connect', 'View details', and 'Instance settings', which is highlighted with a red box. Below the table, the instance details for 'i-0186058675d5e7667 (k21 Windows - Server)' are visible, with tabs for 'Details', 'Security', 'Networking', 'Storage', 'Status checks', 'Monitoring', and 'Tags'.

2. Now click on Change Termination Protection



This screenshot shows the 'Instance settings' dropdown menu for the selected instance. The option 'Change termination protection' is highlighted with a red box and a red arrow. Other options in the menu include 'Attach to Auto Scaling Group', 'Change shutdown behavior', 'Change auto-recovery behavior', 'Change instance type', 'Change Nitro Enclaves', 'Change credit specification', 'Change resource based naming options', 'Modify instance placement', 'Modify Capacity Reservation settings', 'Edit user data', 'Allow tags in instance metadata', and 'Manage tags'.

3. Its disable by default, to enable it please checkmark the box.



The screenshot shows the 'Change termination protection' dialog box. It contains the following information:

- Title:** Change termination protection
- Description:** To prevent your instance from being accidentally terminated, you can enable termination protection for the instance. [Learn more](#)
- Instance ID:** i-027ab584a27b51435 (K21-Windows-Instance)
- Termination protection:** A checkbox labeled 'Enable' which is currently unchecked.
- Buttons:** 'Cancel' and 'Save' buttons at the bottom right.

10 TROUBLESHOOTING

Note: In this section we have shared the solution for the errors that were faced by our trainee's so please check.

10.1 Invalid Access Key Id

10.1.1 ERROR

InvalidAccessKeyId: The AWS Access Key Id you provided does not exist in our records.

```

Command Prompt
C:\Users\User\aws-nodejs-sample>aws configure
AWS Access Key ID [*****ASE6]: *****YAUASE6
AWS Secret Access Key [*****yIRX]: aVghSLMe7726~uL7D~K~h~e~S~7Dy~g~y~IRX
Default region name [us-west-2]:
Default output format [json]:

C:\Users\User\aws-nodejs-sample>node sample.js
InvalidAccessKeyId: The AWS Access Key Id you provided does not exist in our records.
    at Request.extractError (C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\services\s3.js:837:35)
    at Request.callListeners (C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\sequential_executor.js:106:20)
    at Request.emit (C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\sequential_executor.js:78:10)
    at Request.emit (C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\request.js:688:14)
    at Request.transition (C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\request.js:22:10)
    at AcceptorStateMachine.runTo (C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\state_machine.js:14:12)
    at C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\state_machine.js:26:10
    at Request.<anonymous> (C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\request.js:38:9)
    at Request.<anonymous> (C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\request.js:690:12)
    at Request.callListeners (C:\Users\User\aws-nodejs-sample\node_modules\aws-sdk\lib\sequential_executor.js:116:18) {
  code: 'InvalidAccessKeyId',
  region: null,
  time: 2020-09-18T18:49:11.945Z,
  requestId: '93B8DFF0C28BC3F',
  extendedRequestId: 'JYNS3ddpPmMgRLsf/Y1sb9aDhnVnY51ByvzPEvI/R4If5Lthgd35w14mXJVis9BZEVOiM01dEyQ=',
  cfId: undefined,
  statusCode: 403,
  retryable: false,
  retryDelay: 52.86287578941995
}

C:\Users\User\aws-nodejs-sample>

```

10.1.2 CAUSE

Access key id provided is incorrect or has been removed or replaced for the user.

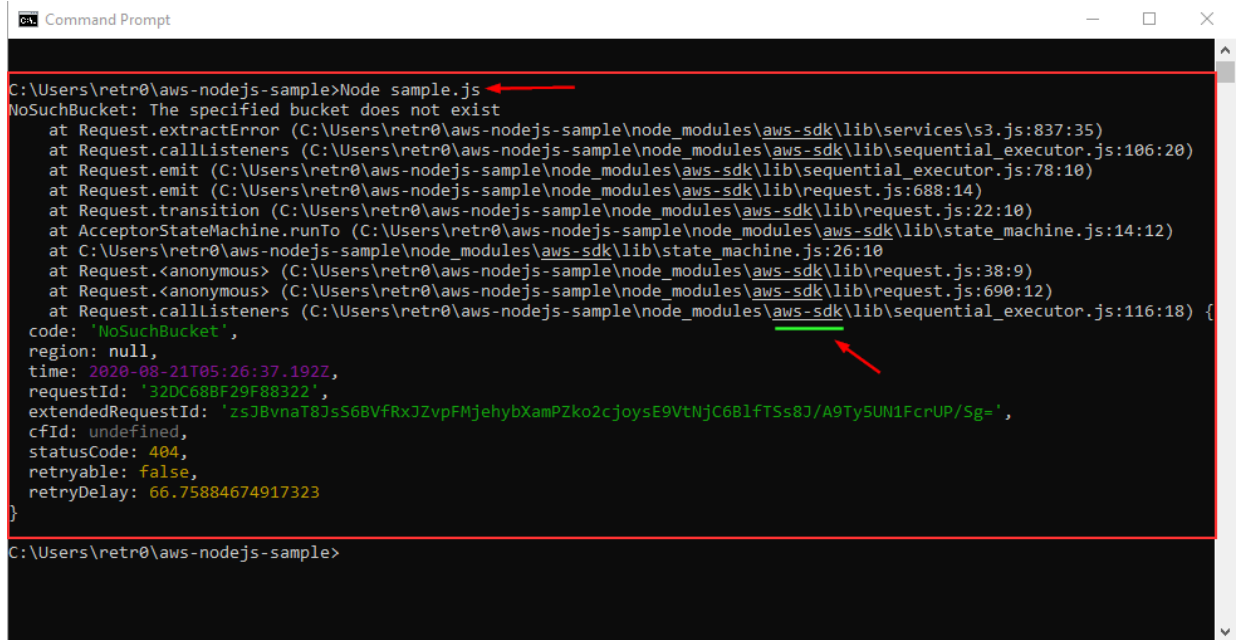
10.1.3 FIX

Provide a correct key or regenerate a new one follow steps to create key from Section: Creating AWS Access Keys.

10.2 No Such Bucket

10.2.1 ERROR

Code: No Such Bucket.



```
Command Prompt
C:\Users\retr0\aws-nodejs-sample>Node sample.js
NoSuchBucket: The specified bucket does not exist
    at Request.extractError (C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\services\s3.js:837:35)
    at Request.callListeners (C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\sequential_executor.js:106:20)
    at Request.emit (C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\sequential_executor.js:78:10)
    at Request.emit (C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\request.js:688:14)
    at Request.transition (C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\request.js:22:10)
    at AcceptorStateMachine.runTo (C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\state_machine.js:14:12)
    at C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\state_machine.js:26:10
    at Request.<anonymous> (C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\request.js:38:9)
    at Request.<anonymous> (C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\request.js:690:12)
    at Request.callListeners (C:\Users\retr0\aws-nodejs-sample\node_modules\aws-sdk\lib\sequential_executor.js:116:18) {
  code: 'NoSuchBucket',
  region: null,
  time: 2020-08-21T05:26:37.192Z,
  requestId: '32DC68BF29F88322',
  extendedRequestId: 'zsJBvnaT8JsS6BVfRxJZvpFMjehybXamPZko2cjoysE9VtNjC6B1fTSs8J/A9Ty5UN1FcrUP/Sg=',
  cfId: undefined,
  statusCode: 404,
  retryable: false,
  retryDelay: 66.75884674917323
}
C:\Users\retr0\aws-nodejs-sample>
```

10.2.2 CAUSE

Default region name entered incorrectly

10.2.3 FIX

While configuring AWS using Access key and Secret key ID you need to enter the default region name as us-east-1

10.3 Set New Password and Generate New Secret Key for Existing IAM User

10.3.1 ERROR

Not able to access the IAM User

10.3.2 CAUSE

Forget to download user security credentials (.CSV file)

Add user

1 2 3 4 5

✓ **Success**

You successfully created the users shown below. You can view and download user security credentials. You can also email users instructions for signing in to the AWS Management Console. This is the last time these credentials will be available to download. However, you can create new credentials at any time.

Users with AWS Management Console access can sign-in at: <https://343596716223.signin.aws.amazon.com/console>

Download .csv

Forget to download CSV file

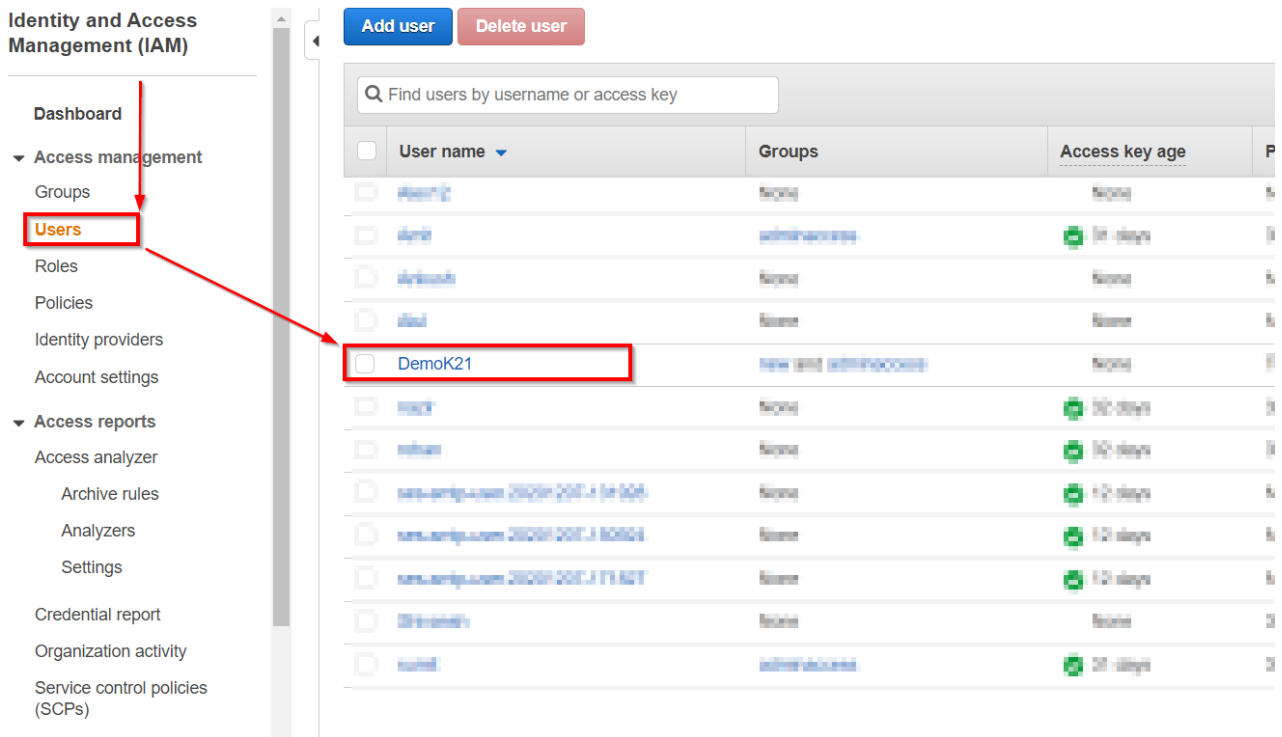
	User	Access key ID	Secret access key	Password	Email login instructions
▶	✓ DemoK21	AKIAU77TV2C7SNCHAK6A	***** Show	***** Show	Send email ↗

Close

10.3.3 FIX

Following are the steps to download New **Sign-in credentials** and **Access keys**

Step 1: Open the IAM console. Click **Users** in the **Details** pane, click the appropriate IAM user.



Identity and Access Management (IAM)

Dashboard

▼ Access management

- Groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings

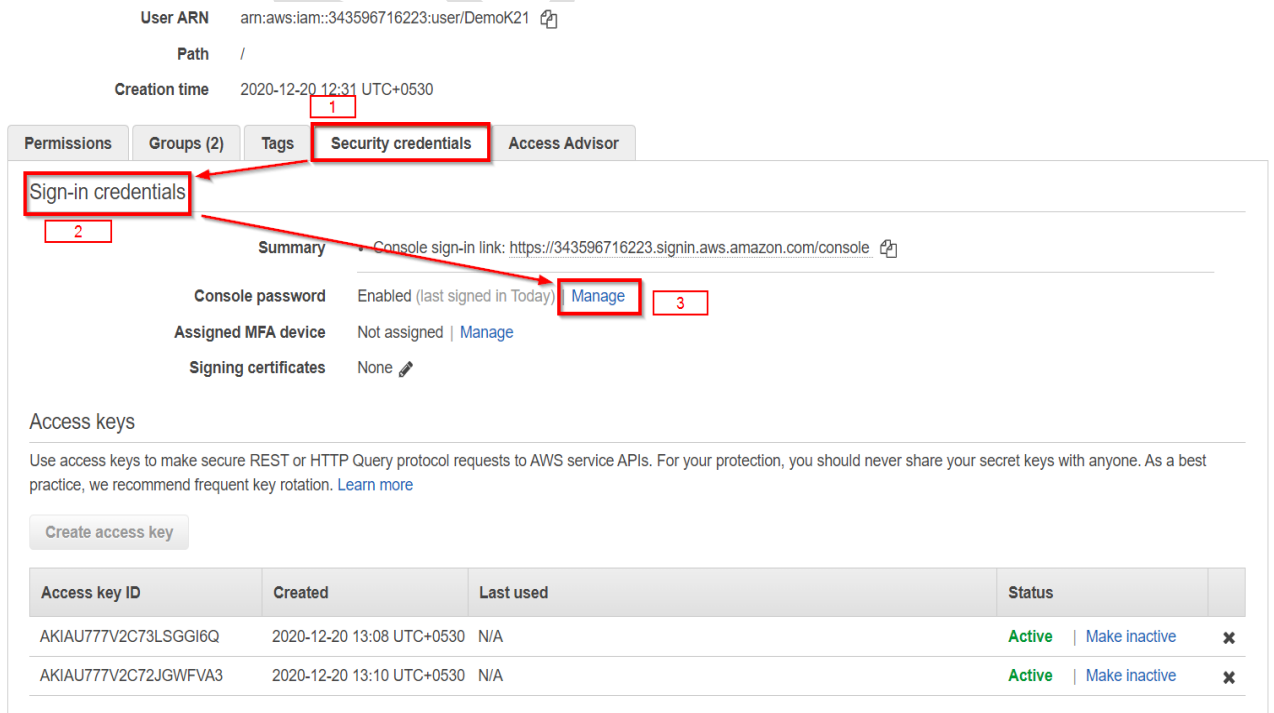
▼ Access reports

- Access analyzer
- Archive rules
- Analizers
- Settings
- Credential report
- Organization activity
- Service control policies (SCPs)

Find users by username or access key

☐	User name	Groups	Access key age	P
☐	admin	None	None	1
☐	admin	adminaccess	14 days	1
☐	admin	None	None	1
☐	admin	None	None	1
☐	Demok21	None and adminaccess	None	1
☐	admin	None	14 days	1
☐	admin	None	14 days	1
☐	aws-iam-user-2020-12-20-13:08:00	None	12 days	1
☐	aws-iam-user-2020-12-20-13:08:00	None	12 days	1
☐	aws-iam-user-2020-12-20-13:08:00	None	12 days	1
☐	aws-iam-user-2020-12-20-13:08:00	None	12 days	1
☐	admin	None	None	1
☐	admin	adminaccess	14 days	1

Step2: Select **Security Credentials**→**Sign-in credentials** click on **Manage**.



User ARN: arn:aws:iam::343596716223:user/Demok21

Path: /

Creation time: 2020-12-20 12:31 UTC+0530

Permissions Groups (2) Tags **Security credentials** Access Advisor

Sign-in credentials

Summary

- Console sign-in link: <https://343596716223.signin.aws.amazon.com/console>
- Console password: Enabled (last signed in Today) **Manage**
- Assigned MFA device: Not assigned | [Manage](#)
- Signing certificates: None

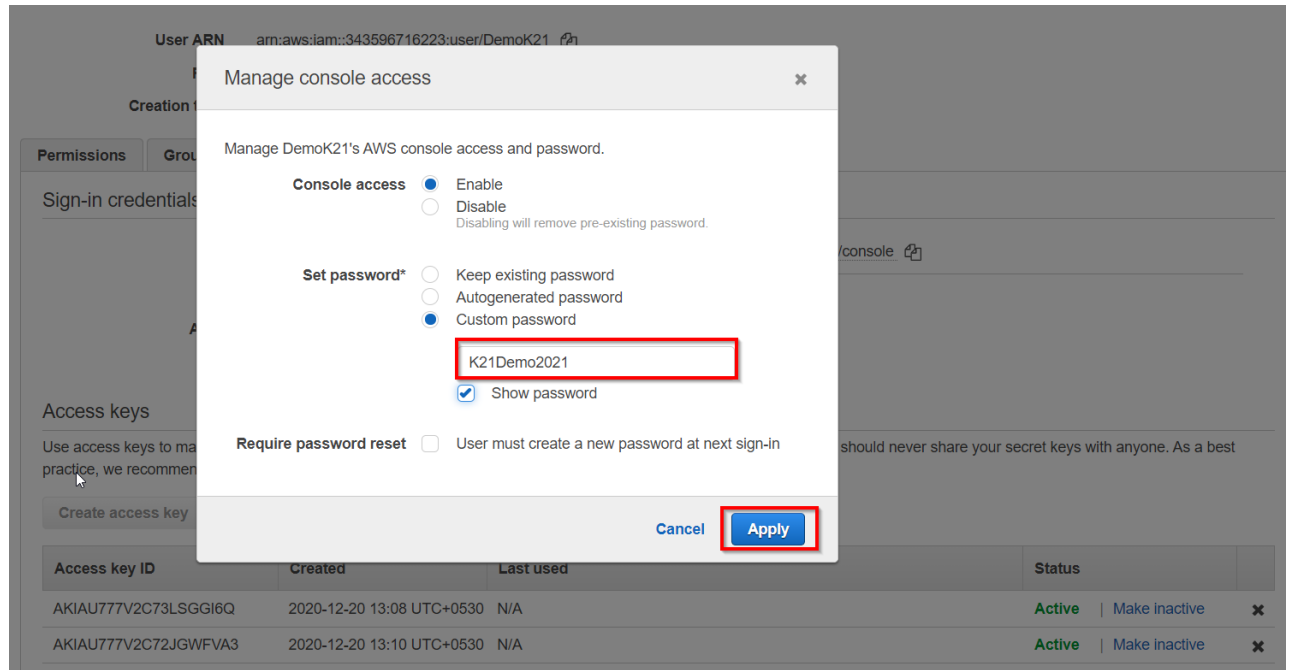
Access keys

Use access keys to make secure REST or HTTP Query protocol requests to AWS service APIs. For your protection, you should never share your secret keys with anyone. As a best practice, we recommend frequent key rotation. [Learn more](#)

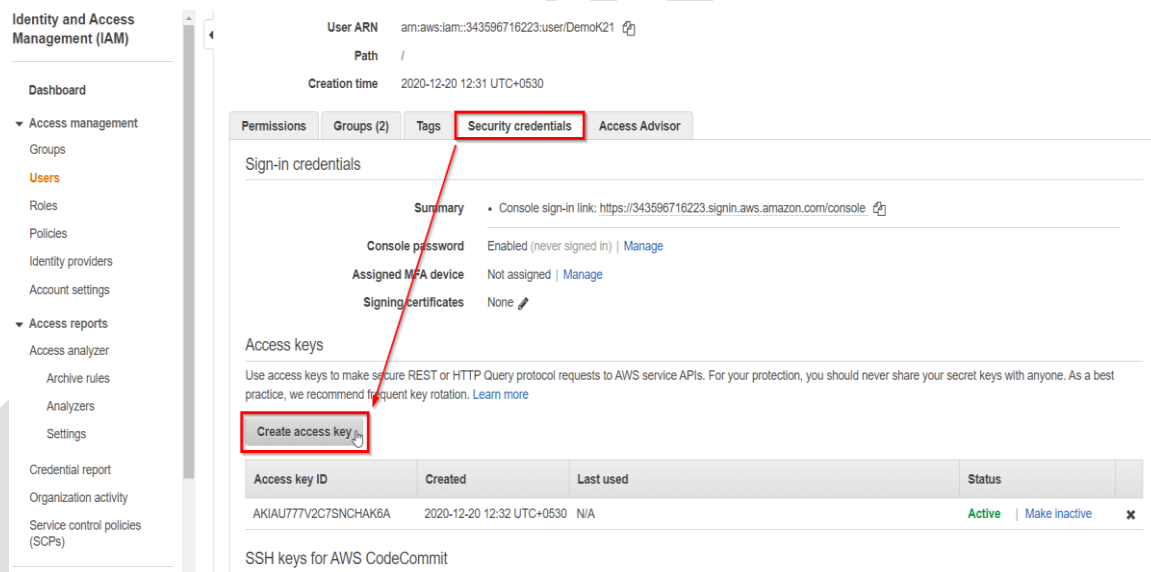
Create access key

Access key ID	Created	Last used	Status
AKIAU77V2C73LSGGI6Q	2020-12-20 13:08 UTC+0530	N/A	Active Make inactive ✕
AKIAU77V2C73JGWFVA3	2020-12-20 13:10 UTC+0530	N/A	Active Make inactive ✕

Step3: Select Custom password then click on Apply

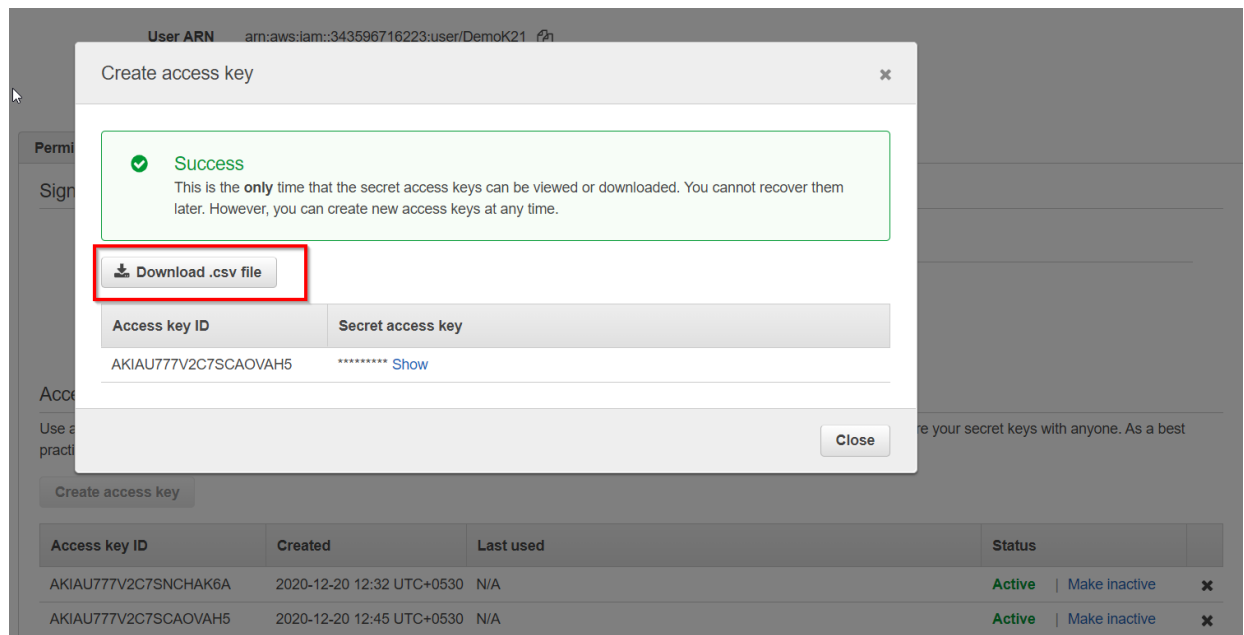


Step 4: To generate new Access key Click Create Access Key on the Security Credentials tab.



Note: If you already have the maximum of two access keys—active or inactive—you must delete one first before proceeding. If you need more than two root access keys, IAM users (each of whom can be assigned their own access keys) would probably better suit your requirements.

Step 5: Download the newly created credentials, when prompted to do so in the key creation wizard.



Step 6: Make your unused access keys inactive in case you need to roll back, and then delete them when you are sure that they're no longer needed (after you have confirmed that they are not in use by any of your applications). The access key last used feature can help you validate if keys are still in use.

Note: Access keys in an "Inactive" state still count toward your maximum of two access keys at any given time.

10.4 Unable to install AWS CLI on MAC

10.4.1 ERROR

aws --version command not found

```

Last login: Sat Dec 26 13:52:14 on ttys000
vitaliy@MacBook-Pro ~ % aws --version
zsh: command not found: aws
vitaliy@MacBook-Pro ~ % sudo ./awscli-bundle/install -i /usr/local/aws -b /usr/local/bin/aws
Password:
Running cmd: /System/Library/Frameworks/Python.framework/Versions/2.7/Resources/Python.app/Contents/MacOS/Python virtualenv.py --no-download
--python /System/Library/Frameworks/Python.framework/Versions/2.7/Resources/Python.app/Contents/MacOS/Python /usr/local/aws
Traceback (most recent call last):
  File "./awscli-bundle/install", line 247, in <module>
    main()
  File "./awscli-bundle/install", line 223, in main
    create_install_structure(working_dir, opts.install_dir)
  File "./awscli-bundle/install", line 88, in create_install_structure
    create_virtualenv(location=install_dir, working_dir=working_dir)
  File "./awscli-bundle/install", line 113, in _create_virtualenv_external
    location))
  File "./awscli-bundle/install", line 68, in run
    p.returncode, cmd, output))
__main__.BadRCError: Bad rc (100) for cmd '/System/Library/Frameworks/Python.framework/Versions/2.7/Resources/Python.app/Contents/MacOS/Pyth
on virtualenv.py --no-download --python /System/Library/Frameworks/Python.framework/Versions/2.7/Resources/Python.app/Contents/MacOS/Python
/usr/local/aws': Already using interpreter /System/Library/Frameworks/Python.framework/Versions/2.7/Resources/Python.app/Contents/MacOS/Pyth
on
New python executable in /usr/local/aws/bin/python
ERROR: The executable /usr/local/aws/bin/python is not functioning
ERROR: It thinks sys.prefix is '/private/tmp/tmpNgox5K/virtualenv-16.7.8' (should be u'/usr/local/aws')
ERROR: virtualenv is not compatible with this system or executable

vitaliy@MacBook-Pro ~ % aws --version
zsh: command not found: aws
vitaliy@MacBook-Pro ~ %

```

10.4.2 CAUSE

AWS CLI is not installed on your mac compute/laptop

10.4.3 FIX

For latest version of AWSCLI for **MAC** download from this link -

<https://awscli.amazonaws.com/AWSCLIV2.pkg>

Step: Install the AWS CLI using the curl command as shown in the following screenshot.

```

vitaliy@MacBook-Pro ~ % curl "https://awscli.amazonaws.com/AWSCLIV2.pkg" -o "AWSCLIV2.pkg"
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
           Dload Upload   Total   Spent    Left  Speed
100 22.4M  100 22.4M    0     0 5639k      0  0:00:04  0:00:04 --:--:-- 5639k
vitaliy@MacBook-Pro ~ % sudo installer -pkg AWSCLIV2.pkg -target /
installer: Package name is AWS Command Line Interface
installer: Installing at base path /
installer: The install was successful.
vitaliy@MacBook-Pro ~ % aws --version
aws-cli/2.1.14 Python/3.7.4 Darwin/20.2.0 exe/x86_64 prompt/off
vitaliy@MacBook-Pro ~ %

```

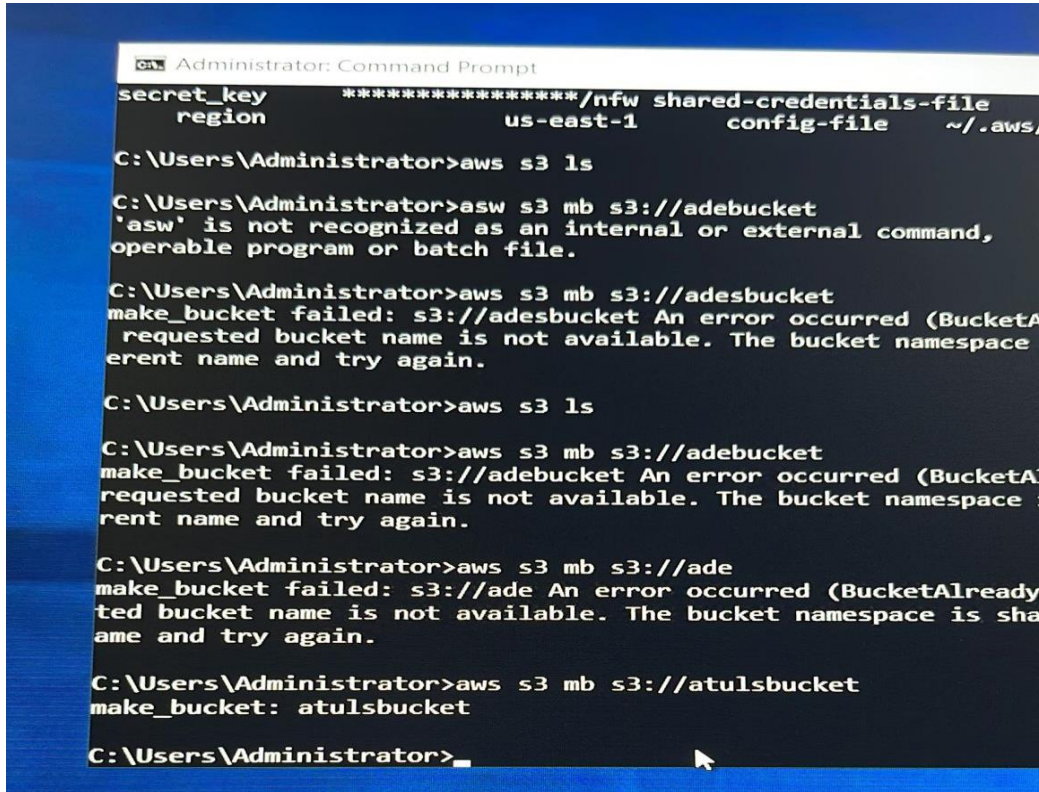
For more information check this user guide for installing AWS CLI on MAC –

<https://docs.aws.amazon.com/cli/latest/userguide/install-cliv2-mac.html>

10.5 Unable to create a Bucket

10.5.1 ERROR

Bucket name is already taken



```
Administrator: Command Prompt
secret_key *****/nfw shared-credentials-file
region us-east-1 config-file ~/.aws/

C:\Users\Administrator>aws s3 ls

C:\Users\Administrator>aws s3 mb s3://adebucket
'aws' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Administrator>aws s3 mb s3://adesbucket
make_bucket failed: s3://adesbucket An error occurred (BucketAlreadyExists) when calling the CreateBucket operation: requested bucket name is not available. The bucket namespace is shared by all users in the specified Amazon S3 region. Please use a different name and try again.

C:\Users\Administrator>aws s3 ls

C:\Users\Administrator>aws s3 mb s3://adebucket
make_bucket failed: s3://adebucket An error occurred (BucketAlreadyExists) when calling the CreateBucket operation: requested bucket name is not available. The bucket namespace is shared by all users in the specified Amazon S3 region. Please use a different name and try again.

C:\Users\Administrator>aws s3 mb s3://ade
make_bucket failed: s3://ade An error occurred (BucketAlreadyExists) when calling the CreateBucket operation: requested bucket name is not available. The bucket namespace is shared by all users in the specified Amazon S3 region. Please use a different name and try again.

C:\Users\Administrator>aws s3 mb s3://atulsbucket
make_bucket: atulsbucket

C:\Users\Administrator>
```

10.5.2 ISSUE

The bucket name is already taken by someone else.

10.5.3 FIX

Try with a unique name globally. So, make sure you follow all the rules before naming the S3 bucket:

- Bucket names must be between 3 and 63 characters long.
- Bucket names can consist only of lowercase letters, numbers, dot (.), and hyphens (-).
- Bucket names must begin and end with a letter or number.

10.6 Error: "AccessDenied" error in AWS S3

10.6.1 ERROR

Access Denied.

```
[ec2-user@ip-172-31-48-17 ~]$ aws s3 mb s3://aws-devops-course-sameer-new --region us-east-1
make_bucket failed: s3://aws-devops-course-sameer-new An error occurred (AccessDenied) when calling the CreateBucket operation: Access Denied
```

10.6.2 CAUSE

Insufficient permission.

10.6.3 FIX

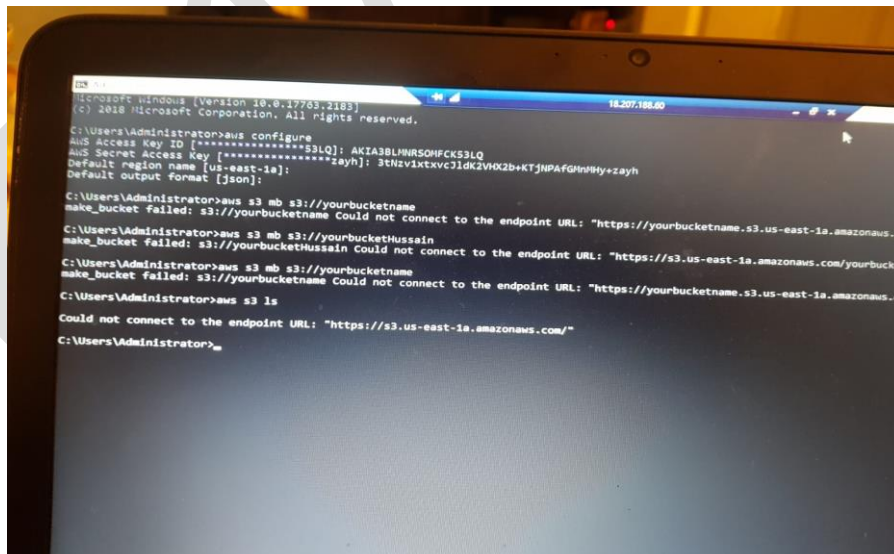
To resolve this issue, you should check the permissions of the IAM user or role that you are using to create the bucket. The user or role must have the necessary permissions to create an S3 bucket. Check if the group in which IAM user is present has **AdministratorAccess** policy attached to it.

You can check the permissions by navigating to the IAM console, selecting the user or role in question, and reviewing the attached policies.

10.7 Unable to access S3

10.7.1 ERROR

Could not connect to S3 Endpoint.



10.7.2 CAUSE

You don't have sufficient permissions and You have selected us-east-1a in region.

10.7.3 FIX

- 1) You have to select us-east-1 in region.
- 2) If you are accessing it via IAM user, make sure the user has appropriate permission to access and create the bucket.

10.8 Not able to delete Bucket via CLI

10.8.1 ERROR

Not able to delete the bucket.

```
C:\Users\Administrator>aws s3 rb s3://node-sdk-sample-d80d1160-9f42-4218-962e-b4f5a9c21416
remove_bucket failed: s3://node-sdk-sample-d80d1160-9f42-4218-962e-b4f5a9c21416 An error occurred (BucketNotEmpty) when
calling the DeleteBucket operation: The bucket you tried to delete is not empty
```

10.8.2 CAUSE

If you are unable to delete the bucket via CLI or facing an error "Bucket is not empty".

10.8.3 FIX

Step1: Delete Objects

```
aws s3 rm s3://yourbucketname --recursive
```

Step2: Delete s3 bucket

```
aws s3 rb s3://yourbucketname
```

Step3: Cross checked in command prompt and GUI

```
aws s3 ls
```

11 SUMMARY

This activity guide cover steps for:

1. Installing AWS CLI On Windows EC2 Instance
2. Configuring AWS CLI
3. Create S3 Bucket using AWS CLI
4. Terminate Resources